

SIMULATION IDEAS

The model is assumed to be

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \approx \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \quad (1)$$

where matrix $\mathbf{X}_{ij}^* := (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R \times S}$ with $\alpha_{rs}(t) := \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \rangle$. Within one patient, the eyes are correlated by

$$\begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \sigma^2(t) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right).$$

Case 1: 2D Tensor Covariate

DGP. The tensor size is set to be $D = 2$, $p_1 = p_2 = 60$, and split by $p'_1 = p'_2 = 12$ such that by $S^{(1)} = S^{(2)} = 5$. Set the rank $R = 4$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathbf{N}(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\begin{aligned} \alpha_{1rs}(t) &= 5\sqrt{\frac{rs}{RS}}(t - 0.2)^2, \quad \alpha_{2rs}(t) = \sqrt{\frac{rs}{RS}} \left(e^{-(3t-1)^2} - 0.75 \right), \quad \alpha_{3rs}(t) = \sqrt{\frac{rs}{RS}} \sin(2\pi(t - 0.5)), \\ \alpha_{4rs}(t) &= \sqrt{\frac{rs}{RS}} \left[0.45 \cdot 5(t - 0.2)^2 + 0.35(e^{-(3t-1)^2} - 0.75) + 0.20 \sin(2\pi(t - 0.5)) \right], \\ \alpha_{5rs}(t) &= \sqrt{\frac{rs}{RS}} \left[1.10 \exp \left\{ -\frac{(t - 0.30)^2}{2(0.08)^2} \right\} - 0.95 \exp \left\{ -\frac{(t - 0.72)^2}{2(0.11)^2} \right\} \right], \\ \alpha_{6rs}(t) &= \sqrt{\frac{rs}{RS}} \left[18(t - 0.2)(t - 0.55)(t - 0.85) - 0.05 \right] \end{aligned}$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (2.0, 1.0, -1.0, 0.5)^\top$. We will vary $\sigma^2(t)$ to 4 subcases as Case 1(a)-(d).

$$\sigma_0^2(t) \equiv 1, \quad \sigma_1^2(t) = 1 + 0.3 \sin(2\pi t), \quad \sigma_2^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t), \quad \text{and} \quad \sigma_3^2(t) = 0.5 + 0.5 \sin^2(\pi t). \quad (2)$$

Estimation Steps. (1) We treat the paired data as iid, and get initial estimate $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\boldsymbol{\beta}}^\dagger$. (2) Then estimate covariance structure $\hat{\sigma}^2(t)$ by eq(15)-(16) in AOAS draft. (3) Get final estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\boldsymbol{\beta}}^*$ from eq(14)-(15) of AOAS draft.

Simulation Results. We vary $n = 1000/2000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{4rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Table 1: MAE errors ($\times 10^{-2}$) for Case 1(a) - $\sigma_0(t) \equiv 1$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.06 ($\pm$ 0.17)	5.08 ($\pm$ 0.18)	2.09 ($\pm$ 0.78)	2.12 ($\pm$ 0.77)	30.32 ($\pm$ 2.66)	2.81 ($\pm$ 1.91)
		0.3	5.05 ($\pm$ 0.20)	4.94 ($\pm$ 0.18)	2.44 ($\pm$ 1.01)	2.49 ($\pm$ 1.07)	30.23 ($\pm$ 3.48)	7.48 ($\pm$ 3.06)
		0.6	5.06 ($\pm$ 0.24)	4.53 ($\pm$ 0.17)	2.66 ($\pm$ 1.10)	2.69 ($\pm$ 1.13)	30.19 ($\pm$ 3.71)	15.09 ($\pm$ 2.55)
		0.9	5.06 ($\pm$ 0.27)	3.57 ($\pm$ 0.18)	2.88 ($\pm$ 1.13)	2.84 ($\pm$ 1.15)	30.14 ($\pm$ 3.99)	22.70 ($\pm$ 1.92)
	2000	0.0	3.29 ($\pm$ 0.10)	3.30 ($\pm$ 0.10)	1.25 ($\pm$ 0.43)	1.24 ($\pm$ 0.44)	13.31 ($\pm$ 1.88)	1.37 ($\pm$ 0.80)
		0.3	3.32 ($\pm$ 0.11)	3.22 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.36 ($\pm$ 0.61)	13.24 ($\pm$ 1.82)	4.12 ($\pm$ 1.64)
		0.6	3.33 ($\pm$ 0.11)	2.89 ($\pm$ 0.10)	1.53 ($\pm$ 0.65)	1.47 ($\pm$ 0.67)	13.15 ($\pm$ 2.04)	8.48 ($\pm$ 1.41)
		0.9	3.33 ($\pm$ 0.11)	2.16 ($\pm$ 0.07)	1.68 ($\pm$ 0.70)	1.56 ($\pm$ 0.72)	13.07 ($\pm$ 2.32)	12.92 ($\pm$ 1.03)
$\mathbf{A}_2(t)$	1000	0.0	5.08 ($\pm$ 0.21)	5.10 ($\pm$ 0.21)	2.09 ($\pm$ 0.79)	2.12 ($\pm$ 0.78)	30.33 ($\pm$ 2.72)	2.94 ($\pm$ 2.11)
		0.3	5.03 ($\pm$ 0.18)	4.92 ($\pm$ 0.16)	2.44 ($\pm$ 0.98)	2.51 ($\pm$ 1.03)	30.28 ($\pm$ 3.56)	6.87 ($\pm$ 2.92)
		0.6	5.03 ($\pm$ 0.20)	4.50 ($\pm$ 0.16)	2.66 ($\pm$ 1.08)	2.71 ($\pm$ 1.11)	30.25 ($\pm$ 3.78)	14.46 ($\pm$ 2.31)
		0.9	5.02 ($\pm$ 0.22)	3.52 ($\pm$ 0.15)	2.89 ($\pm$ 1.12)	2.84 ($\pm$ 1.14)	30.20 ($\pm$ 4.05)	22.09 ($\pm$ 1.48)
	2000	0.0	3.28 ($\pm$ 0.08)	3.29 ($\pm$ 0.08)	1.21 ($\pm$ 0.46)	1.21 ($\pm$ 0.46)	13.43 ($\pm$ 1.89)	1.56 ($\pm$ 0.96)
		0.3	3.29 ($\pm$ 0.12)	3.19 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.37 ($\pm$ 0.63)	13.34 ($\pm$ 1.78)	3.87 ($\pm$ 1.59)
		0.6	3.29 ($\pm$ 0.12)	2.86 ($\pm$ 0.10)	1.54 ($\pm$ 0.64)	1.48 ($\pm$ 0.70)	13.24 ($\pm$ 2.03)	8.22 ($\pm$ 1.31)
		0.9	3.29 ($\pm$ 0.11)	2.09 ($\pm$ 0.08)	1.69 ($\pm$ 0.69)	1.60 ($\pm$ 0.74)	13.12 ($\pm$ 2.34)	12.64 ($\pm$ 0.98)
$\mathbf{A}_3(t)$	1000	0.0	6.49 ($\pm$ 0.22)	6.60 ($\pm$ 0.23)	2.21 ($\pm$ 0.76)	2.27 ($\pm$ 0.76)	17.16 ($\pm$ 2.64)	2.85 ($\pm$ 2.21)
		0.3	6.52 ($\pm$ 0.18)	6.56 ($\pm$ 0.17)	2.46 ($\pm$ 1.11)	2.53 ($\pm$ 1.09)	17.08 ($\pm$ 3.21)	11.00 ($\pm$ 3.08)
		0.6	6.53 ($\pm$ 0.19)	6.29 ($\pm$ 0.16)	2.70 ($\pm$ 1.18)	2.73 ($\pm$ 1.16)	17.15 ($\pm$ 3.26)	22.32 ($\pm$ 2.77)
		0.9	6.54 ($\pm$ 0.22)	5.75 ($\pm$ 0.15)	2.95 ($\pm$ 1.21)	2.89 ($\pm$ 1.22)	17.24 ($\pm$ 3.35)	33.66 ($\pm$ 2.46)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.49 ($\pm$ 0.56)	1.48 ($\pm$ 0.58)	9.82 ($\pm$ 1.17)	1.33 ($\pm$ 0.98)
		0.3	5.05 ($\pm$ 0.12)	5.09 ($\pm$ 0.12)	1.41 ($\pm$ 0.59)	1.38 ($\pm$ 0.61)	10.29 ($\pm$ 1.30)	8.32 ($\pm$ 1.90)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.55 ($\pm$ 0.65)	1.49 ($\pm$ 0.66)	10.41 ($\pm$ 1.36)	16.81 ($\pm$ 1.83)
		0.9	5.06 ($\pm$ 0.13)	4.50 ($\pm$ 0.09)	1.67 ($\pm$ 0.70)	1.58 ($\pm$ 0.70)	10.57 ($\pm$ 1.48)	25.32 ($\pm$ 1.67)
$\mathbf{A}_4(t)$	1000	0.0	4.87 ($\pm$ 0.19)	4.88 ($\pm$ 0.20)	2.09 ($\pm$ 0.76)	2.13 ($\pm$ 0.75)	31.77 ($\pm$ 2.69)	2.85 ($\pm$ 1.97)
		0.3	4.85 ($\pm$ 0.18)	4.73 ($\pm$ 0.16)	2.43 ($\pm$ 1.01)	2.47 ($\pm$ 1.07)	31.68 ($\pm$ 3.47)	6.83 ($\pm$ 2.97)
		0.6	4.85 ($\pm$ 0.21)	4.28 ($\pm$ 0.15)	2.65 ($\pm$ 1.10)	2.67 ($\pm$ 1.14)	31.64 ($\pm$ 3.69)	13.98 ($\pm$ 2.37)
		0.9	4.86 ($\pm$ 0.24)	3.20 ($\pm$ 0.16)	2.88 ($\pm$ 1.13)	2.82 ($\pm$ 1.15)	31.59 ($\pm$ 3.97)	21.14 ($\pm$ 1.64)
	2000	0.0	3.02 ($\pm$ 0.09)	3.02 ($\pm$ 0.09)	1.26 ($\pm$ 0.45)	1.26 ($\pm$ 0.45)	15.12 ($\pm$ 1.87)	1.44 ($\pm$ 0.78)
		0.3	3.03 ($\pm$ 0.09)	2.91 ($\pm$ 0.09)	1.36 ($\pm$ 0.58)	1.33 ($\pm$ 0.60)	15.04 ($\pm$ 1.85)	3.57 ($\pm$ 1.45)
		0.6	3.04 ($\pm$ 0.09)	2.54 ($\pm$ 0.08)	1.51 ($\pm$ 0.64)	1.45 ($\pm$ 0.66)	14.95 ($\pm$ 2.06)	7.33 ($\pm$ 1.30)
		0.9	3.04 ($\pm$ 0.09)	1.63 ($\pm$ 0.06)	1.65 ($\pm$ 0.69)	1.54 ($\pm$ 0.71)	14.85 ($\pm$ 2.35)	11.21 ($\pm$ 0.88)
$\mathbf{A}_5(t)$	1000	0.0	7.88 ($\pm$ 0.22)	8.19 ($\pm$ 0.24)	2.56 ($\pm$ 1.05)	2.49 ($\pm$ 1.00)	28.32 ($\pm$ 2.37)	3.50 ($\pm$ 2.04)
		0.3	7.84 ($\pm$ 0.26)	8.09 ($\pm$ 0.26)	2.63 ($\pm$ 0.83)	2.71 ($\pm$ 0.96)	28.28 ($\pm$ 2.54)	14.55 ($\pm$ 3.56)
		0.6	7.85 ($\pm$ 0.27)	7.89 ($\pm$ 0.26)	2.85 ($\pm$ 0.88)	2.93 ($\pm$ 1.03)	28.21 ($\pm$ 2.58)	29.92 ($\pm$ 3.58)
		0.9	7.85 ($\pm$ 0.30)	7.48 ($\pm$ 0.28)	3.08 ($\pm$ 0.93)	3.13 ($\pm$ 1.11)	28.19 ($\pm$ 2.66)	45.30 ($\pm$ 3.57)
	2000	0.0	6.46 ($\pm$ 0.14)	6.77 ($\pm$ 0.14)	1.59 ($\pm$ 0.55)	1.53 ($\pm$ 0.50)	38.31 ($\pm$ 2.77)	2.00 ($\pm$ 1.43)
		0.3	6.48 ($\pm$ 0.13)	6.73 ($\pm$ 0.13)	1.66 ($\pm$ 0.63)	1.61 ($\pm$ 0.55)	37.98 ($\pm$ 2.08)	13.04 ($\pm$ 2.74)
		0.6	6.49 ($\pm$ 0.13)	6.57 ($\pm$ 0.13)	1.79 ($\pm$ 0.67)	1.74 ($\pm$ 0.59)	38.02 ($\pm$ 2.04)	26.38 ($\pm$ 2.59)
		0.9	6.48 ($\pm$ 0.13)	6.24 ($\pm$ 0.13)	1.93 ($\pm$ 0.71)	1.85 ($\pm$ 0.61)	38.11 ($\pm$ 2.09)	39.79 ($\pm$ 2.32)
$\mathbf{A}_6(t)$	1000	0.0	6.18 ($\pm$ 0.21)	6.26 ($\pm$ 0.21)	2.10 ($\pm$ 0.84)	2.15 ($\pm$ 0.81)	20.74 ($\pm$ 2.87)	2.91 ($\pm$ 2.08)
		0.3	6.13 ($\pm$ 0.21)	6.12 ($\pm$ 0.20)	2.62 ($\pm$ 1.01)	2.64 ($\pm$ 1.05)	20.63 ($\pm$ 3.74)	9.38 ($\pm$ 3.13)
		0.6	6.12 ($\pm$ 0.23)	5.81 ($\pm$ 0.20)	2.83 ($\pm$ 1.12)	2.82 ($\pm$ 1.13)	20.62 ($\pm$ 3.94)	19.66 ($\pm$ 2.69)
		0.9	6.11 ($\pm$ 0.25)	5.18 ($\pm$ 0.20)	3.04 ($\pm$ 1.19)	2.96 ($\pm$ 1.18)	20.62 ($\pm$ 4.18)	29.96 ($\pm$ 2.07)
	2000	0.0	4.59 ($\pm$ 0.10)	4.65 ($\pm$ 0.10)	1.17 ($\pm$ 0.45)	1.17 ($\pm$ 0.45)	6.90 ($\pm$ 1.09)	1.67 ($\pm$ 1.00)
		0.3	4.61 ($\pm$ 0.13)	4.61 ($\pm$ 0.12)	1.55 ($\pm$ 0.64)	1.54 ($\pm$ 0.67)	7.11 ($\pm$ 1.22)	6.86 ($\pm$ 1.85)
		0.6	4.61 ($\pm$ 0.13)	4.40 ($\pm$ 0.11)	1.71 ($\pm$ 0.69)	1.66 ($\pm$ 0.74)	7.16 ($\pm$ 1.31)	14.36 ($\pm$ 1.69)
		0.9	4.60 ($\pm$ 0.14)	3.95 ($\pm$ 0.08)	1.86 ($\pm$ 0.73)	1.79 ($\pm$ 0.78)	7.26 ($\pm$ 1.44)	21.94 ($\pm$ 1.52)

Table 2: MAE errors ($\times 10^{-2}$) for Case 1(b) - $\sigma_1(t) = 1 + 0.3 \sin(2\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.05 ($\pm$ 0.18)	5.05 ($\pm$ 0.18)	2.12 ($\pm$ 0.80)	2.05 ($\pm$ 0.75)	30.43 ($\pm$ 2.75)	2.93 ($\pm$ 1.94)
		0.3	5.03 ($\pm$ 0.20)	4.92 ($\pm$ 0.18)	2.41 ($\pm$ 1.03)	2.45 ($\pm$ 1.04)	30.27 ($\pm$ 3.47)	7.63 ($\pm$ 3.19)
		0.6	5.03 ($\pm$ 0.23)	4.51 ($\pm$ 0.17)	2.63 ($\pm$ 1.10)	2.65 ($\pm$ 1.10)	30.26 ($\pm$ 3.76)	15.23 ($\pm$ 2.67)
		0.9	5.04 ($\pm$ 0.27)	3.56 ($\pm$ 0.18)	2.86 ($\pm$ 1.14)	2.79 ($\pm$ 1.12)	30.23 ($\pm$ 4.12)	22.80 ($\pm$ 2.02)
	2000	0.0	3.29 ($\pm$ 0.10)	3.29 ($\pm$ 0.10)	1.24 ($\pm$ 0.46)	1.21 ($\pm$ 0.42)	13.47 ($\pm$ 1.92)	1.60 ($\pm$ 0.90)
		0.3	3.31 ($\pm$ 0.11)	3.20 ($\pm$ 0.11)	1.39 ($\pm$ 0.61)	1.34 ($\pm$ 0.58)	13.41 ($\pm$ 1.93)	4.31 ($\pm$ 1.55)
		0.6	3.32 ($\pm$ 0.11)	2.88 ($\pm$ 0.10)	1.53 ($\pm$ 0.67)	1.44 ($\pm$ 0.65)	13.46 ($\pm$ 2.10)	8.63 ($\pm$ 1.31)
		0.9	3.32 ($\pm$ 0.11)	2.15 ($\pm$ 0.07)	1.67 ($\pm$ 0.72)	1.53 ($\pm$ 0.69)	13.57 ($\pm$ 2.25)	12.97 ($\pm$ 0.98)
$\mathbf{A}_2(t)$	1000	0.0	5.05 ($\pm$ 0.21)	5.07 ($\pm$ 0.21)	2.11 ($\pm$ 0.82)	2.06 ($\pm$ 0.75)	30.43 ($\pm$ 2.82)	3.14 ($\pm$ 2.03)
		0.3	4.99 ($\pm$ 0.18)	4.89 ($\pm$ 0.17)	2.42 ($\pm$ 1.01)	2.44 ($\pm$ 1.02)	30.32 ($\pm$ 3.54)	6.97 ($\pm$ 3.07)
		0.6	4.99 ($\pm$ 0.20)	4.47 ($\pm$ 0.16)	2.66 ($\pm$ 1.09)	2.63 ($\pm$ 1.11)	30.32 ($\pm$ 3.83)	14.55 ($\pm$ 2.42)
		0.9	4.99 ($\pm$ 0.22)	3.49 ($\pm$ 0.15)	2.88 ($\pm$ 1.15)	2.78 ($\pm$ 1.12)	30.30 ($\pm$ 4.18)	22.12 ($\pm$ 1.51)
	2000	0.0	3.26 ($\pm$ 0.08)	3.27 ($\pm$ 0.08)	1.19 ($\pm$ 0.46)	1.18 ($\pm$ 0.44)	13.55 ($\pm$ 1.99)	1.77 ($\pm$ 1.12)
		0.3	3.26 ($\pm$ 0.12)	3.17 ($\pm$ 0.12)	1.38 ($\pm$ 0.62)	1.35 ($\pm$ 0.58)	13.41 ($\pm$ 1.94)	4.06 ($\pm$ 1.53)
		0.6	3.26 ($\pm$ 0.12)	2.84 ($\pm$ 0.10)	1.54 ($\pm$ 0.67)	1.46 ($\pm$ 0.65)	13.41 ($\pm$ 2.12)	8.36 ($\pm$ 1.28)
		0.9	3.26 ($\pm$ 0.11)	2.07 ($\pm$ 0.08)	1.69 ($\pm$ 0.71)	1.56 ($\pm$ 0.70)	13.45 ($\pm$ 2.32)	12.69 ($\pm$ 1.00)
$\mathbf{A}_3(t)$	1000	0.0	6.48 ($\pm$ 0.22)	6.59 ($\pm$ 0.23)	2.22 ($\pm$ 0.79)	2.21 ($\pm$ 0.75)	19.24 ($\pm$ 2.34)	2.95 ($\pm$ 2.18)
		0.3	6.51 ($\pm$ 0.18)	6.54 ($\pm$ 0.17)	2.43 ($\pm$ 1.08)	2.49 ($\pm$ 1.08)	19.03 ($\pm$ 2.81)	11.20 ($\pm$ 3.26)
		0.6	6.52 ($\pm$ 0.20)	6.28 ($\pm$ 0.17)	2.67 ($\pm$ 1.15)	2.69 ($\pm$ 1.15)	19.03 ($\pm$ 3.09)	22.52 ($\pm$ 2.99)
		0.9	6.53 ($\pm$ 0.22)	5.74 ($\pm$ 0.16)	2.91 ($\pm$ 1.19)	2.86 ($\pm$ 1.20)	19.06 ($\pm$ 3.45)	33.85 ($\pm$ 2.68)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.52 ($\pm$ 0.55)	1.47 ($\pm$ 0.56)	10.30 ($\pm$ 1.13)	1.39 ($\pm$ 1.04)
		0.3	5.05 ($\pm$ 0.12)	5.08 ($\pm$ 0.12)	1.41 ($\pm$ 0.58)	1.35 ($\pm$ 0.61)	10.84 ($\pm$ 1.29)	8.51 ($\pm$ 1.89)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.54 ($\pm$ 0.63)	1.46 ($\pm$ 0.66)	10.99 ($\pm$ 1.33)	16.96 ($\pm$ 1.81)
		0.9	5.06 ($\pm$ 0.13)	4.49 ($\pm$ 0.09)	1.67 ($\pm$ 0.68)	1.56 ($\pm$ 0.70)	11.22 ($\pm$ 1.37)	25.42 ($\pm$ 1.66)
$\mathbf{A}_4(t)$	1000	0.0	4.85 ($\pm$ 0.20)	4.86 ($\pm$ 0.20)	2.12 ($\pm$ 0.78)	2.05 ($\pm$ 0.71)	31.88 ($\pm$ 2.79)	2.95 ($\pm$ 2.01)
		0.3	4.82 ($\pm$ 0.18)	4.70 ($\pm$ 0.16)	2.39 ($\pm$ 1.03)	2.42 ($\pm$ 1.05)	31.72 ($\pm$ 3.45)	6.98 ($\pm$ 3.13)
		0.6	4.82 ($\pm$ 0.21)	4.26 ($\pm$ 0.15)	2.62 ($\pm$ 1.10)	2.61 ($\pm$ 1.12)	31.71 ($\pm$ 3.75)	14.11 ($\pm$ 2.52)
		0.9	4.82 ($\pm$ 0.24)	3.18 ($\pm$ 0.15)	2.85 ($\pm$ 1.14)	2.77 ($\pm$ 1.13)	31.69 ($\pm$ 4.10)	21.23 ($\pm$ 1.76)
	2000	0.0	3.00 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.25 ($\pm$ 0.47)	1.23 ($\pm$ 0.43)	15.24 ($\pm$ 1.94)	1.65 ($\pm$ 0.92)
		0.3	3.01 ($\pm$ 0.09)	2.89 ($\pm$ 0.09)	1.36 ($\pm$ 0.59)	1.31 ($\pm$ 0.57)	15.15 ($\pm$ 1.94)	3.71 ($\pm$ 1.47)
		0.6	3.02 ($\pm$ 0.09)	2.52 ($\pm$ 0.08)	1.51 ($\pm$ 0.65)	1.42 ($\pm$ 0.64)	15.14 ($\pm$ 2.11)	7.47 ($\pm$ 1.21)
		0.9	3.02 ($\pm$ 0.09)	1.62 ($\pm$ 0.06)	1.65 ($\pm$ 0.70)	1.51 ($\pm$ 0.68)	15.15 ($\pm$ 2.33)	11.25 ($\pm$ 0.85)
$\mathbf{A}_5(t)$	1000	0.0	7.85 ($\pm$ 0.23)	8.15 ($\pm$ 0.25)	2.57 ($\pm$ 1.07)	2.44 ($\pm$ 0.94)	26.70 ($\pm$ 2.27)	3.61 ($\pm$ 2.24)
		0.3	7.81 ($\pm$ 0.25)	8.05 ($\pm$ 0.25)	2.63 ($\pm$ 0.82)	2.60 ($\pm$ 0.94)	26.69 ($\pm$ 2.51)	14.59 ($\pm$ 3.55)
		0.6	7.81 ($\pm$ 0.26)	7.85 ($\pm$ 0.25)	2.85 ($\pm$ 0.86)	2.82 ($\pm$ 1.02)	26.63 ($\pm$ 2.58)	29.96 ($\pm$ 3.56)
		0.9	7.81 ($\pm$ 0.28)	7.45 ($\pm$ 0.27)	3.08 ($\pm$ 0.90)	3.02 ($\pm$ 1.08)	26.58 ($\pm$ 2.69)	45.32 ($\pm$ 3.51)
	2000	0.0	6.44 ($\pm$ 0.13)	6.74 ($\pm$ 0.14)	1.58 ($\pm$ 0.53)	1.48 ($\pm$ 0.47)	38.26 ($\pm$ 2.98)	2.13 ($\pm$ 1.50)
		0.3	6.46 ($\pm$ 0.13)	6.70 ($\pm$ 0.13)	1.67 ($\pm$ 0.63)	1.57 ($\pm$ 0.56)	37.88 ($\pm$ 2.37)	13.22 ($\pm$ 2.83)
		0.6	6.46 ($\pm$ 0.13)	6.54 ($\pm$ 0.13)	1.80 ($\pm$ 0.67)	1.69 ($\pm$ 0.60)	37.88 ($\pm$ 2.32)	26.55 ($\pm$ 2.68)
		0.9	6.46 ($\pm$ 0.13)	6.22 ($\pm$ 0.13)	1.93 ($\pm$ 0.71)	1.80 ($\pm$ 0.62)	37.90 ($\pm$ 2.35)	39.91 ($\pm$ 2.40)
$\mathbf{A}_6(t)$	1000	0.0	6.16 ($\pm$ 0.21)	6.23 ($\pm$ 0.22)	2.11 ($\pm$ 0.87)	2.11 ($\pm$ 0.79)	20.88 ($\pm$ 2.96)	3.10 ($\pm$ 2.08)
		0.3	6.10 ($\pm$ 0.21)	6.09 ($\pm$ 0.20)	2.62 ($\pm$ 1.05)	2.57 ($\pm$ 1.01)	20.76 ($\pm$ 3.59)	9.42 ($\pm$ 3.30)
		0.6	6.09 ($\pm$ 0.23)	5.78 ($\pm$ 0.20)	2.83 ($\pm$ 1.15)	2.75 ($\pm$ 1.09)	20.79 ($\pm$ 3.83)	19.70 ($\pm$ 2.82)
		0.9	6.08 ($\pm$ 0.24)	5.16 ($\pm$ 0.20)	3.04 ($\pm$ 1.23)	2.88 ($\pm$ 1.15)	20.80 ($\pm$ 4.15)	29.97 ($\pm$ 2.13)
	2000	0.0	4.58 ($\pm$ 0.10)	4.64 ($\pm$ 0.10)	1.16 ($\pm$ 0.45)	1.16 ($\pm$ 0.44)	6.98 ($\pm$ 1.29)	1.80 ($\pm$ 1.19)
		0.3	4.58 ($\pm$ 0.13)	4.59 ($\pm$ 0.12)	1.53 ($\pm$ 0.69)	1.52 ($\pm$ 0.64)	7.08 ($\pm$ 1.29)	7.02 ($\pm$ 1.88)
		0.6	4.59 ($\pm$ 0.13)	4.38 ($\pm$ 0.11)	1.68 ($\pm$ 0.75)	1.64 ($\pm$ 0.70)	7.21 ($\pm$ 1.49)	14.49 ($\pm$ 1.74)
		0.9	4.58 ($\pm$ 0.14)	3.94 ($\pm$ 0.08)	1.84 ($\pm$ 0.79)	1.75 ($\pm$ 0.74)	7.38 ($\pm$ 1.73)	22.00 ($\pm$ 1.61)

Table 3: MAE errors ($\times 10^{-2}$) for Case 1(c) - $\sigma_2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	4.28 ($\pm$ 0.14)	4.29 ($\pm$ 0.14)	1.82 ($\pm$ 0.68)	1.78 ($\pm$ 0.66)	20.98 ($\pm$ 2.10)	2.79 ($\pm$ 1.96)
		0.3	4.28 ($\pm$ 0.18)	4.19 ($\pm$ 0.16)	2.08 ($\pm$ 0.85)	2.12 ($\pm$ 0.89)	20.91 ($\pm$ 2.62)	7.39 ($\pm$ 3.17)
		0.6	4.28 ($\pm$ 0.20)	3.84 ($\pm$ 0.15)	2.27 ($\pm$ 0.92)	2.29 ($\pm$ 0.94)	20.89 ($\pm$ 2.80)	14.77 ($\pm$ 2.63)
		0.9	4.28 ($\pm$ 0.23)	3.06 ($\pm$ 0.15)	2.45 ($\pm$ 0.96)	2.40 ($\pm$ 0.96)	20.85 ($\pm$ 3.03)	22.15 ($\pm$ 1.96)
	2000	0.0	2.85 ($\pm$ 0.09)	2.86 ($\pm$ 0.09)	1.09 ($\pm$ 0.38)	1.04 ($\pm$ 0.38)	8.56 ($\pm$ 1.44)	1.51 ($\pm$ 0.91)
		0.3	2.87 ($\pm$ 0.09)	2.79 ($\pm$ 0.09)	1.23 ($\pm$ 0.52)	1.13 ($\pm$ 0.51)	8.47 ($\pm$ 1.43)	4.17 ($\pm$ 1.73)
		0.6	2.88 ($\pm$ 0.09)	2.53 ($\pm$ 0.09)	1.35 ($\pm$ 0.57)	1.23 ($\pm$ 0.56)	8.41 ($\pm$ 1.58)	8.51 ($\pm$ 1.42)
		0.9	2.88 ($\pm$ 0.10)	1.96 ($\pm$ 0.06)	1.48 ($\pm$ 0.61)	1.31 ($\pm$ 0.60)	8.37 ($\pm$ 1.77)	12.91 ($\pm$ 1.01)
$\mathbf{A}_2(t)$	1000	0.0	4.29 ($\pm$ 0.17)	4.31 ($\pm$ 0.17)	1.83 ($\pm$ 0.69)	1.77 ($\pm$ 0.67)	21.00 ($\pm$ 2.15)	3.03 ($\pm$ 2.04)
		0.3	4.24 ($\pm$ 0.15)	4.16 ($\pm$ 0.14)	2.08 ($\pm$ 0.83)	2.13 ($\pm$ 0.87)	20.97 ($\pm$ 2.70)	6.68 ($\pm$ 2.98)
		0.6	4.24 ($\pm$ 0.17)	3.81 ($\pm$ 0.13)	2.27 ($\pm$ 0.89)	2.30 ($\pm$ 0.93)	20.95 ($\pm$ 2.87)	14.04 ($\pm$ 2.36)
		0.9	4.23 ($\pm$ 0.18)	3.00 ($\pm$ 0.13)	2.48 ($\pm$ 0.92)	2.41 ($\pm$ 0.97)	20.91 ($\pm$ 3.09)	21.43 ($\pm$ 1.49)
	2000	0.0	2.83 ($\pm$ 0.07)	2.84 ($\pm$ 0.07)	1.04 ($\pm$ 0.40)	1.01 ($\pm$ 0.39)	8.68 ($\pm$ 1.45)	1.75 ($\pm$ 1.15)
		0.3	2.83 ($\pm$ 0.10)	2.76 ($\pm$ 0.10)	1.22 ($\pm$ 0.53)	1.15 ($\pm$ 0.51)	8.56 ($\pm$ 1.41)	3.87 ($\pm$ 1.72)
		0.6	2.83 ($\pm$ 0.10)	2.49 ($\pm$ 0.08)	1.36 ($\pm$ 0.56)	1.25 ($\pm$ 0.58)	8.49 ($\pm$ 1.58)	8.19 ($\pm$ 1.38)
		0.9	2.83 ($\pm$ 0.10)	1.87 ($\pm$ 0.06)	1.49 ($\pm$ 0.59)	1.34 ($\pm$ 0.61)	8.42 ($\pm$ 1.81)	12.56 ($\pm$ 0.99)
$\mathbf{A}_3(t)$	1000	0.0	5.84 ($\pm$ 0.19)	5.96 ($\pm$ 0.20)	1.94 ($\pm$ 0.68)	1.95 ($\pm$ 0.68)	9.43 ($\pm$ 1.33)	3.00 ($\pm$ 2.14)
		0.3	5.88 ($\pm$ 0.15)	5.94 ($\pm$ 0.15)	2.12 ($\pm$ 0.98)	2.15 ($\pm$ 0.90)	9.57 ($\pm$ 1.61)	11.74 ($\pm$ 3.09)
		0.6	5.88 ($\pm$ 0.16)	5.74 ($\pm$ 0.14)	2.32 ($\pm$ 1.05)	2.32 ($\pm$ 0.95)	9.61 ($\pm$ 1.60)	23.73 ($\pm$ 2.84)
		0.9	5.89 ($\pm$ 0.17)	5.34 ($\pm$ 0.13)	2.54 ($\pm$ 1.06)	2.45 ($\pm$ 1.00)	9.69 ($\pm$ 1.68)	35.74 ($\pm$ 2.62)
	2000	0.0	4.72 ($\pm$ 0.10)	4.84 ($\pm$ 0.11)	1.36 ($\pm$ 0.51)	1.29 ($\pm$ 0.51)	10.46 ($\pm$ 0.95)	1.38 ($\pm$ 1.05)
		0.3	4.72 ($\pm$ 0.11)	4.80 ($\pm$ 0.10)	1.28 ($\pm$ 0.50)	1.17 ($\pm$ 0.52)	10.69 ($\pm$ 1.10)	9.40 ($\pm$ 2.06)
		0.6	4.73 ($\pm$ 0.12)	4.65 ($\pm$ 0.10)	1.39 ($\pm$ 0.55)	1.26 ($\pm$ 0.56)	10.78 ($\pm$ 1.15)	18.86 ($\pm$ 1.97)
		0.9	4.73 ($\pm$ 0.12)	4.37 ($\pm$ 0.08)	1.50 ($\pm$ 0.59)	1.34 ($\pm$ 0.60)	10.92 ($\pm$ 1.25)	28.35 ($\pm$ 1.83)
$\mathbf{A}_4(t)$	1000	0.0	4.07 ($\pm$ 0.15)	4.07 ($\pm$ 0.16)	1.81 ($\pm$ 0.67)	1.79 ($\pm$ 0.64)	22.43 ($\pm$ 2.13)	2.89 ($\pm$ 1.95)
		0.3	4.05 ($\pm$ 0.15)	3.95 ($\pm$ 0.14)	2.07 ($\pm$ 0.86)	2.09 ($\pm$ 0.89)	22.37 ($\pm$ 2.61)	6.57 ($\pm$ 3.04)
		0.6	4.06 ($\pm$ 0.18)	3.57 ($\pm$ 0.13)	2.26 ($\pm$ 0.93)	2.27 ($\pm$ 0.94)	22.34 ($\pm$ 2.78)	13.34 ($\pm$ 2.41)
		0.9	4.05 ($\pm$ 0.20)	2.65 ($\pm$ 0.13)	2.46 ($\pm$ 0.96)	2.38 ($\pm$ 0.96)	22.31 ($\pm$ 3.01)	20.13 ($\pm$ 1.62)
	2000	0.0	2.55 ($\pm$ 0.08)	2.55 ($\pm$ 0.08)	1.10 ($\pm$ 0.40)	1.05 ($\pm$ 0.38)	10.37 ($\pm$ 1.44)	1.56 ($\pm$ 0.90)
		0.3	2.56 ($\pm$ 0.08)	2.45 ($\pm$ 0.08)	1.19 ($\pm$ 0.50)	1.11 ($\pm$ 0.50)	10.27 ($\pm$ 1.46)	3.45 ($\pm$ 1.54)
		0.6	2.56 ($\pm$ 0.08)	2.15 ($\pm$ 0.07)	1.33 ($\pm$ 0.55)	1.21 ($\pm$ 0.55)	10.21 ($\pm$ 1.61)	7.01 ($\pm$ 1.30)
		0.9	2.56 ($\pm$ 0.08)	1.39 ($\pm$ 0.05)	1.45 ($\pm$ 0.59)	1.28 ($\pm$ 0.59)	10.14 ($\pm$ 1.82)	10.67 ($\pm$ 0.83)
$\mathbf{A}_5(t)$	1000	0.0	7.99 ($\pm$ 0.23)	8.31 ($\pm$ 0.25)	2.56 ($\pm$ 1.06)	2.50 ($\pm$ 1.00)	32.41 ($\pm$ 2.33)	3.50 ($\pm$ 2.03)
		0.3	7.96 ($\pm$ 0.26)	8.21 ($\pm$ 0.26)	2.64 ($\pm$ 0.83)	2.69 ($\pm$ 0.96)	32.40 ($\pm$ 2.47)	14.82 ($\pm$ 3.62)
		0.6	7.96 ($\pm$ 0.28)	8.00 ($\pm$ 0.27)	2.87 ($\pm$ 0.90)	2.92 ($\pm$ 1.03)	32.35 ($\pm$ 2.51)	30.51 ($\pm$ 3.66)
		0.9	7.97 ($\pm$ 0.30)	7.60 ($\pm$ 0.29)	3.09 ($\pm$ 0.96)	3.12 ($\pm$ 1.09)	32.35 ($\pm$ 2.59)	46.21 ($\pm$ 3.67)
	2000	0.0	6.52 ($\pm$ 0.13)	6.83 ($\pm$ 0.14)	1.58 ($\pm$ 0.56)	1.53 ($\pm$ 0.49)	40.76 ($\pm$ 2.52)	1.90 ($\pm$ 1.35)
		0.3	6.55 ($\pm$ 0.13)	6.80 ($\pm$ 0.13)	1.67 ($\pm$ 0.62)	1.61 ($\pm$ 0.58)	40.52 ($\pm$ 1.85)	13.24 ($\pm$ 2.74)
		0.6	6.55 ($\pm$ 0.13)	6.63 ($\pm$ 0.13)	1.80 ($\pm$ 0.65)	1.74 ($\pm$ 0.62)	40.56 ($\pm$ 1.81)	26.77 ($\pm$ 2.60)
		0.9	6.55 ($\pm$ 0.14)	6.29 ($\pm$ 0.13)	1.93 ($\pm$ 0.69)	1.85 ($\pm$ 0.64)	40.62 ($\pm$ 1.89)	40.36 ($\pm$ 2.34)
$\mathbf{A}_6(t)$	1000	0.0	6.26 ($\pm$ 0.22)	6.34 ($\pm$ 0.22)	2.08 ($\pm$ 0.86)	2.14 ($\pm$ 0.78)	22.13 ($\pm$ 2.88)	3.02 ($\pm$ 2.21)
		0.3	6.20 ($\pm$ 0.21)	6.19 ($\pm$ 0.20)	2.64 ($\pm$ 1.05)	2.55 ($\pm$ 1.01)	21.97 ($\pm$ 3.67)	9.59 ($\pm$ 3.29)
		0.6	6.19 ($\pm$ 0.23)	5.88 ($\pm$ 0.20)	2.86 ($\pm$ 1.15)	2.73 ($\pm$ 1.10)	21.94 ($\pm$ 3.87)	20.22 ($\pm$ 2.83)
		0.9	6.18 ($\pm$ 0.26)	5.26 ($\pm$ 0.21)	3.07 ($\pm$ 1.24)	2.88 ($\pm$ 1.14)	21.92 ($\pm$ 4.10)	30.87 ($\pm$ 2.19)
	2000	0.0	4.61 ($\pm$ 0.10)	4.68 ($\pm$ 0.10)	1.18 ($\pm$ 0.46)	1.15 ($\pm$ 0.44)	7.91 ($\pm$ 1.13)	1.58 ($\pm$ 1.01)
		0.3	4.64 ($\pm$ 0.14)	4.64 ($\pm$ 0.12)	1.54 ($\pm$ 0.64)	1.50 ($\pm$ 0.68)	8.20 ($\pm$ 1.19)	7.03 ($\pm$ 1.92)
		0.6	4.64 ($\pm$ 0.14)	4.42 ($\pm$ 0.11)	1.69 ($\pm$ 0.69)	1.62 ($\pm$ 0.74)	8.24 ($\pm$ 1.26)	14.72 ($\pm$ 1.77)
		0.9	4.63 ($\pm$ 0.14)	3.97 ($\pm$ 0.09)	1.84 ($\pm$ 0.73)	1.74 ($\pm$ 0.78)	8.31 ($\pm$ 1.40)	22.50 ($\pm$ 1.60)

Table 4: MAE errors ($\times 10^{-2}$) for Case 1(d) - $\sigma_3(t) = 0.5 + 0.5 \sin^2(\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.17 ($\pm$ 0.18)	5.18 ($\pm$ 0.19)	2.08 ($\pm$ 0.79)	2.09 ($\pm$ 0.75)	31.65 ($\pm$ 2.68)	2.92 ($\pm$ 2.04)
		0.3	5.15 ($\pm$ 0.21)	5.03 ($\pm$ 0.18)	2.45 ($\pm$ 1.03)	2.44 ($\pm$ 1.05)	31.52 ($\pm$ 3.55)	7.70 ($\pm$ 3.10)
		0.6	5.16 ($\pm$ 0.24)	4.62 ($\pm$ 0.17)	2.68 ($\pm$ 1.10)	2.64 ($\pm$ 1.11)	31.47 ($\pm$ 3.79)	15.65 ($\pm$ 2.60)
		0.9	5.16 ($\pm$ 0.28)	3.67 ($\pm$ 0.19)	2.90 ($\pm$ 1.16)	2.79 ($\pm$ 1.12)	31.40 ($\pm$ 4.05)	23.60 ($\pm$ 2.02)
	2000	0.0	3.35 ($\pm$ 0.10)	3.35 ($\pm$ 0.10)	1.25 ($\pm$ 0.43)	1.23 ($\pm$ 0.43)	14.39 ($\pm$ 1.76)	1.31 ($\pm$ 0.81)
		0.3	3.38 ($\pm$ 0.11)	3.27 ($\pm$ 0.11)	1.38 ($\pm$ 0.59)	1.35 ($\pm$ 0.60)	14.46 ($\pm$ 1.71)	4.27 ($\pm$ 1.69)
		0.6	3.39 ($\pm$ 0.11)	2.94 ($\pm$ 0.10)	1.53 ($\pm$ 0.64)	1.46 ($\pm$ 0.66)	14.38 ($\pm$ 1.97)	8.81 ($\pm$ 1.48)
		0.9	3.39 ($\pm$ 0.11)	2.19 ($\pm$ 0.07)	1.67 ($\pm$ 0.69)	1.55 ($\pm$ 0.70)	14.32 ($\pm$ 2.29)	13.44 ($\pm$ 1.10)
$\mathbf{A}_2(t)$	1000	0.0	5.19 ($\pm$ 0.22)	5.20 ($\pm$ 0.22)	2.08 ($\pm$ 0.80)	2.11 ($\pm$ 0.76)	31.67 ($\pm$ 2.75)	3.07 ($\pm$ 2.20)
		0.3	5.13 ($\pm$ 0.18)	5.02 ($\pm$ 0.17)	2.46 ($\pm$ 1.00)	2.45 ($\pm$ 1.01)	31.56 ($\pm$ 3.61)	7.11 ($\pm$ 3.01)
		0.6	5.13 ($\pm$ 0.20)	4.60 ($\pm$ 0.16)	2.69 ($\pm$ 1.10)	2.65 ($\pm$ 1.10)	31.52 ($\pm$ 3.83)	15.04 ($\pm$ 2.38)
		0.9	5.13 ($\pm$ 0.23)	3.62 ($\pm$ 0.16)	2.92 ($\pm$ 1.16)	2.80 ($\pm$ 1.11)	31.47 ($\pm$ 4.09)	23.00 ($\pm$ 1.58)
	2000	0.0	3.34 ($\pm$ 0.08)	3.33 ($\pm$ 0.08)	1.20 ($\pm$ 0.46)	1.19 ($\pm$ 0.45)	14.82 ($\pm$ 1.71)	1.45 ($\pm$ 0.96)
		0.3	3.36 ($\pm$ 0.12)	3.25 ($\pm$ 0.12)	1.39 ($\pm$ 0.59)	1.35 ($\pm$ 0.63)	14.94 ($\pm$ 1.66)	4.05 ($\pm$ 1.61)
		0.6	3.36 ($\pm$ 0.12)	2.91 ($\pm$ 0.11)	1.54 ($\pm$ 0.64)	1.47 ($\pm$ 0.70)	14.87 ($\pm$ 1.93)	8.58 ($\pm$ 1.36)
		0.9	3.36 ($\pm$ 0.12)	2.12 ($\pm$ 0.08)	1.69 ($\pm$ 0.69)	1.58 ($\pm$ 0.74)	14.79 ($\pm$ 2.25)	13.19 ($\pm$ 1.04)
$\mathbf{A}_3(t)$	1000	0.0	6.60 ($\pm$ 0.23)	6.71 ($\pm$ 0.24)	2.21 ($\pm$ 0.77)	2.23 ($\pm$ 0.74)	19.12 ($\pm$ 2.45)	2.91 ($\pm$ 2.24)
		0.3	6.62 ($\pm$ 0.19)	6.66 ($\pm$ 0.17)	2.48 ($\pm$ 1.09)	2.49 ($\pm$ 1.09)	19.06 ($\pm$ 3.18)	11.24 ($\pm$ 3.14)
		0.6	6.63 ($\pm$ 0.20)	6.39 ($\pm$ 0.16)	2.72 ($\pm$ 1.16)	2.69 ($\pm$ 1.17)	19.13 ($\pm$ 3.23)	22.90 ($\pm$ 2.84)
		0.9	6.64 ($\pm$ 0.22)	5.86 ($\pm$ 0.16)	2.96 ($\pm$ 1.20)	2.85 ($\pm$ 1.23)	19.23 ($\pm$ 3.26)	34.59 ($\pm$ 2.55)
	2000	0.0	5.10 ($\pm$ 0.11)	5.19 ($\pm$ 0.12)	1.48 ($\pm$ 0.55)	1.47 ($\pm$ 0.57)	12.79 ($\pm$ 1.33)	1.30 ($\pm$ 0.97)
		0.3	5.11 ($\pm$ 0.12)	5.14 ($\pm$ 0.12)	1.40 ($\pm$ 0.60)	1.37 ($\pm$ 0.60)	13.23 ($\pm$ 1.41)	8.47 ($\pm$ 1.87)
		0.6	5.12 ($\pm$ 0.13)	4.94 ($\pm$ 0.11)	1.53 ($\pm$ 0.66)	1.48 ($\pm$ 0.64)	13.33 ($\pm$ 1.53)	17.14 ($\pm$ 1.83)
		0.9	5.12 ($\pm$ 0.13)	4.53 ($\pm$ 0.09)	1.65 ($\pm$ 0.72)	1.58 ($\pm$ 0.68)	13.45 ($\pm$ 1.70)	25.86 ($\pm$ 1.70)
$\mathbf{A}_4(t)$	1000	0.0	4.97 ($\pm$ 0.21)	4.98 ($\pm$ 0.21)	2.08 ($\pm$ 0.77)	2.10 ($\pm$ 0.72)	33.11 ($\pm$ 2.70)	2.93 ($\pm$ 2.14)
		0.3	4.94 ($\pm$ 0.19)	4.82 ($\pm$ 0.16)	2.43 ($\pm$ 1.03)	2.42 ($\pm$ 1.04)	32.96 ($\pm$ 3.54)	7.05 ($\pm$ 3.01)
		0.6	4.95 ($\pm$ 0.21)	4.37 ($\pm$ 0.15)	2.67 ($\pm$ 1.11)	2.61 ($\pm$ 1.13)	32.92 ($\pm$ 3.77)	14.53 ($\pm$ 2.42)
		0.9	4.95 ($\pm$ 0.25)	3.30 ($\pm$ 0.16)	2.90 ($\pm$ 1.16)	2.77 ($\pm$ 1.13)	32.86 ($\pm$ 4.03)	22.04 ($\pm$ 1.74)
	2000	0.0	3.06 ($\pm$ 0.09)	3.06 ($\pm$ 0.09)	1.26 ($\pm$ 0.44)	1.24 ($\pm$ 0.45)	16.05 ($\pm$ 1.84)	1.38 ($\pm$ 0.82)
		0.3	3.08 ($\pm$ 0.10)	2.96 ($\pm$ 0.10)	1.35 ($\pm$ 0.58)	1.32 ($\pm$ 0.59)	16.08 ($\pm$ 1.78)	3.71 ($\pm$ 1.48)
		0.6	3.08 ($\pm$ 0.09)	2.59 ($\pm$ 0.08)	1.50 ($\pm$ 0.64)	1.43 ($\pm$ 0.65)	15.99 ($\pm$ 2.05)	7.66 ($\pm$ 1.36)
		0.9	3.09 ($\pm$ 0.09)	1.66 ($\pm$ 0.06)	1.64 ($\pm$ 0.69)	1.53 ($\pm$ 0.69)	15.89 ($\pm$ 2.39)	11.73 ($\pm$ 0.95)
$\mathbf{A}_5(t)$	1000	0.0	7.26 ($\pm$ 0.19)	7.59 ($\pm$ 0.22)	2.34 ($\pm$ 0.93)	2.14 ($\pm$ 0.85)	25.26 ($\pm$ 2.29)	3.63 ($\pm$ 2.11)
		0.3	7.23 ($\pm$ 0.23)	7.52 ($\pm$ 0.24)	2.31 ($\pm$ 0.75)	2.33 ($\pm$ 0.83)	25.11 ($\pm$ 2.60)	15.73 ($\pm$ 3.80)
		0.6	7.23 ($\pm$ 0.25)	7.36 ($\pm$ 0.25)	2.51 ($\pm$ 0.78)	2.51 ($\pm$ 0.88)	25.06 ($\pm$ 2.68)	32.31 ($\pm$ 3.85)
		0.9	7.23 ($\pm$ 0.26)	7.08 ($\pm$ 0.26)	2.69 ($\pm$ 0.80)	2.65 ($\pm$ 0.96)	25.03 ($\pm$ 2.79)	48.89 ($\pm$ 3.85)
	2000	0.0	6.14 ($\pm$ 0.13)	6.49 ($\pm$ 0.14)	1.47 ($\pm$ 0.49)	1.33 ($\pm$ 0.44)	39.37 ($\pm$ 2.68)	2.28 ($\pm$ 1.74)
		0.3	6.16 ($\pm$ 0.12)	6.46 ($\pm$ 0.12)	1.52 ($\pm$ 0.58)	1.39 ($\pm$ 0.45)	38.85 ($\pm$ 2.12)	14.64 ($\pm$ 2.96)
		0.6	6.16 ($\pm$ 0.12)	6.34 ($\pm$ 0.12)	1.63 ($\pm$ 0.62)	1.49 ($\pm$ 0.48)	38.90 ($\pm$ 2.11)	29.59 ($\pm$ 2.81)
		0.9	6.15 ($\pm$ 0.13)	6.12 ($\pm$ 0.13)	1.74 ($\pm$ 0.66)	1.58 ($\pm$ 0.49)	39.00 ($\pm$ 2.17)	44.58 ($\pm$ 2.55)
$\mathbf{A}_6(t)$	1000	0.0	5.57 ($\pm$ 0.18)	5.63 ($\pm$ 0.18)	1.88 ($\pm$ 0.72)	1.85 ($\pm$ 0.69)	12.29 ($\pm$ 1.89)	2.83 ($\pm$ 2.15)
		0.3	5.52 ($\pm$ 0.18)	5.52 ($\pm$ 0.18)	2.27 ($\pm$ 0.86)	2.32 ($\pm$ 0.90)	12.39 ($\pm$ 2.44)	9.89 ($\pm$ 3.12)
		0.6	5.51 ($\pm$ 0.19)	5.28 ($\pm$ 0.18)	2.45 ($\pm$ 0.94)	2.47 ($\pm$ 0.98)	12.43 ($\pm$ 2.53)	20.64 ($\pm$ 2.74)
		0.9	5.49 ($\pm$ 0.21)	4.80 ($\pm$ 0.18)	2.63 ($\pm$ 0.98)	2.58 ($\pm$ 1.03)	12.48 ($\pm$ 2.66)	31.40 ($\pm$ 2.19)
	2000	0.0	4.29 ($\pm$ 0.09)	4.35 ($\pm$ 0.10)	1.04 ($\pm$ 0.40)	1.03 ($\pm$ 0.38)	9.57 ($\pm$ 1.10)	1.83 ($\pm$ 1.18)
		0.3	4.30 ($\pm$ 0.11)	4.32 ($\pm$ 0.10)	1.39 ($\pm$ 0.58)	1.36 ($\pm$ 0.56)	9.76 ($\pm$ 1.20)	7.66 ($\pm$ 1.92)
		0.6	4.30 ($\pm$ 0.11)	4.16 ($\pm$ 0.09)	1.52 ($\pm$ 0.63)	1.47 ($\pm$ 0.61)	9.77 ($\pm$ 1.36)	15.93 ($\pm$ 1.74)
		0.9	4.29 ($\pm$ 0.11)	3.83 ($\pm$ 0.07)	1.65 ($\pm$ 0.67)	1.58 ($\pm$ 0.65)	9.79 ($\pm$ 1.54)	24.27 ($\pm$ 1.57)

Below we present three examples, with fitted-versus-true curves for $\mathbf{A}_2$, $\mathbf{A}_4$, and $\mathbf{A}_6$, respectively. These plots show that our method is effective across a range of different $\mathbf{A}(t)$ configurations.

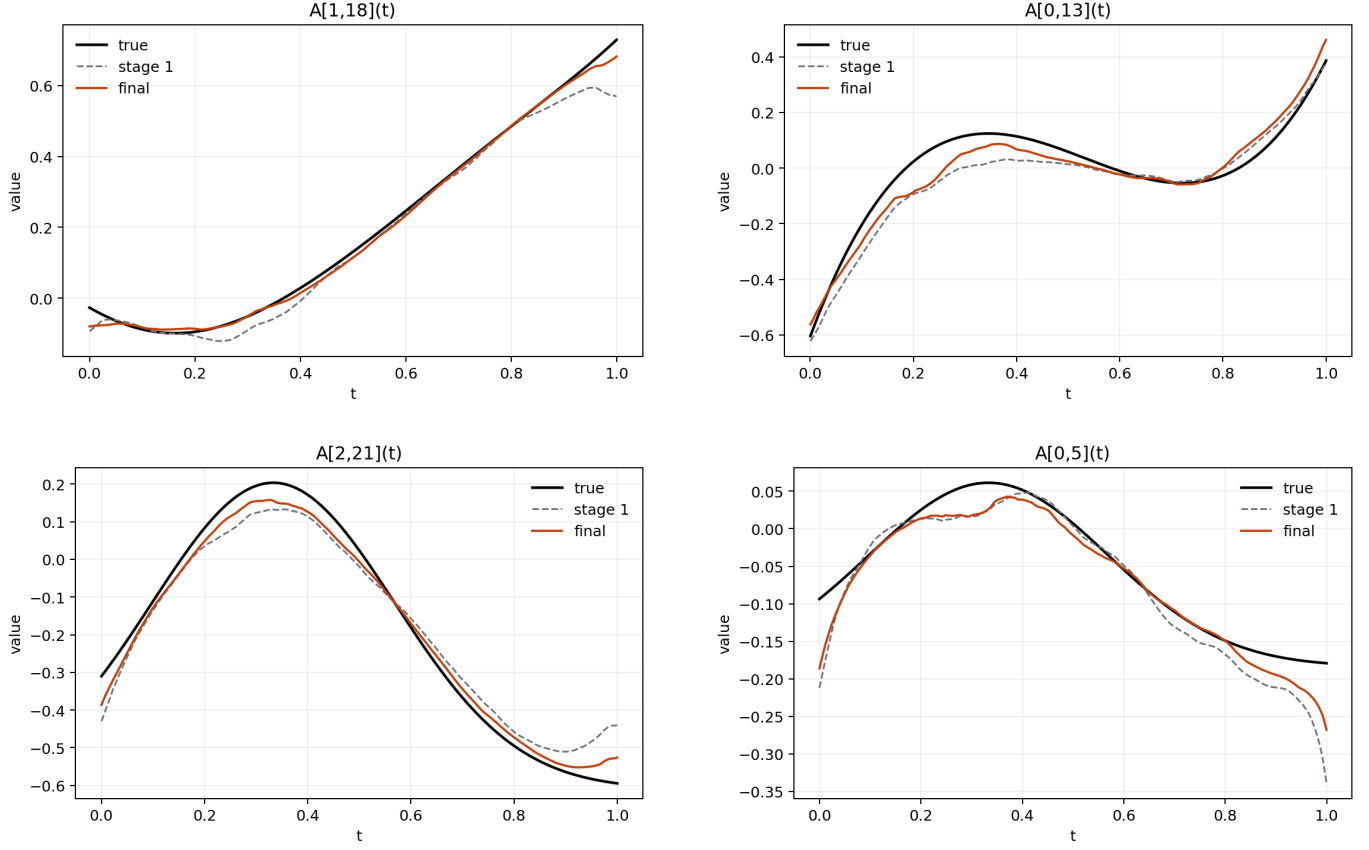


Figure 1: Selected estimated coefficient-function curves from Case 1.

We also present 3 examples of fitted-versus-true curves for $\sigma^2(t)$. They show that our method is effective for estimating the variance function across different settings.

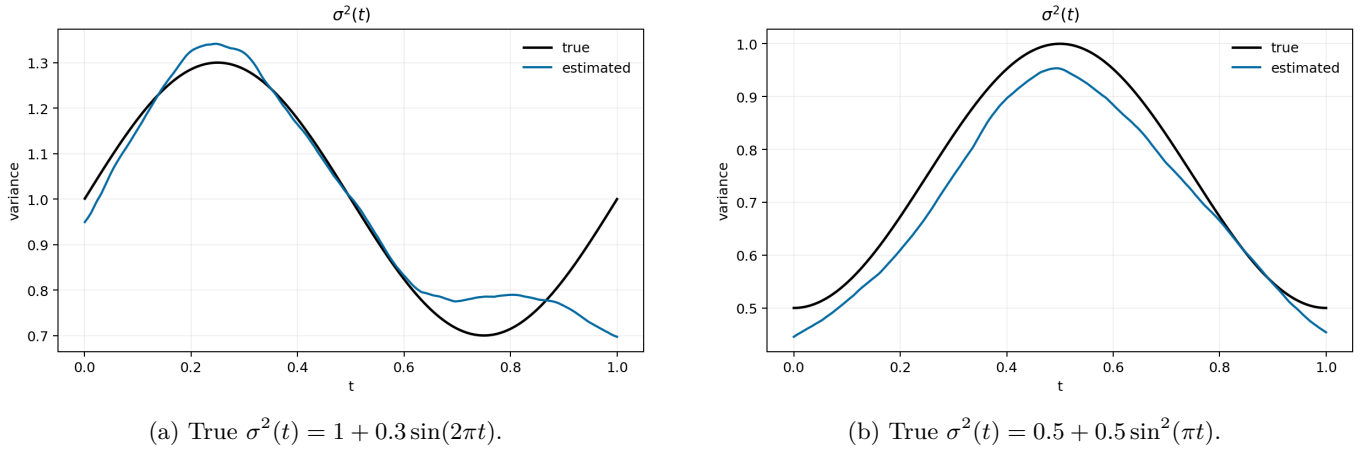


Figure 2: Selected estimated variance-function curves from Case 1.

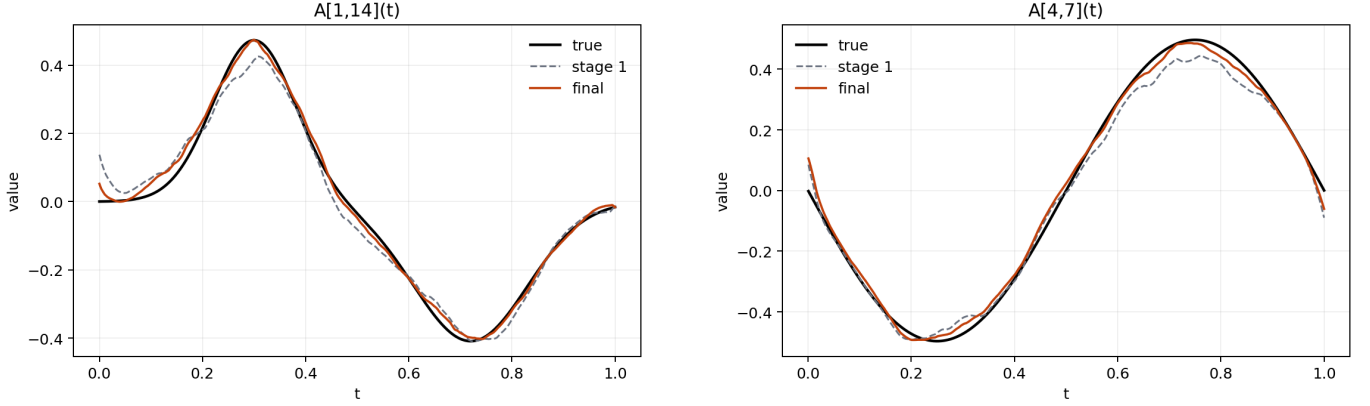
Case 2: 3D Tensor Covariate

DGP. The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 48$, and split by $p'_1 = p'_2 = p'_3 = 16$ such that by $S^{(1)} = S^{(2)} = S^{(3)} = 3$. Set the rank $R = 6$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} N(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$ where $\alpha_{rs}(t)$ has the same 4 choices as in Case 1. And everything else is the same.

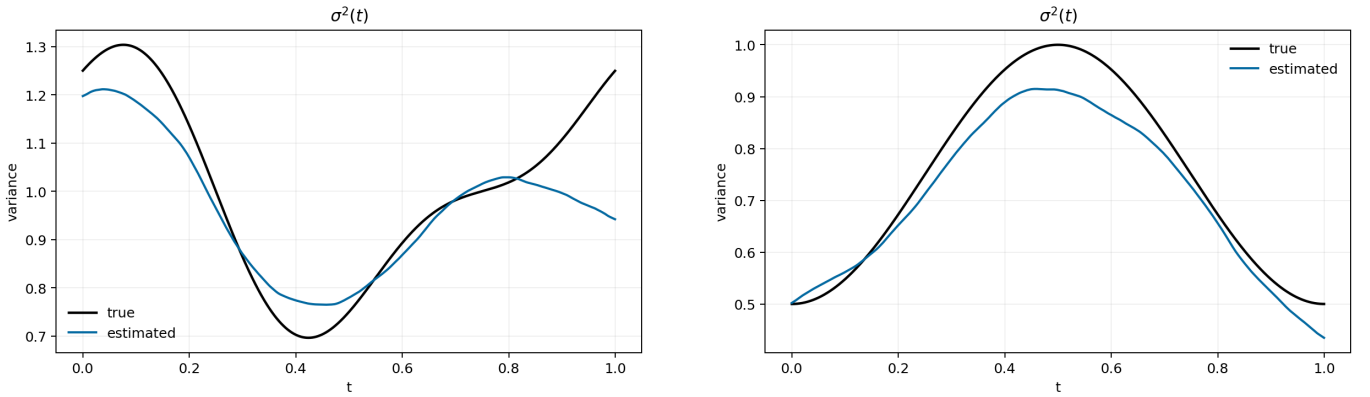
Simulation Results. We vary $n = 2000/5000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{6rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. Tables 5–8 summarize the Case 2 results under the four variance settings. Figure 3 presents representative fitted-versus-true coefficient-function curves under $\mathbf{A}_5(t)$ and $\mathbf{A}_3(t)$, and Figure 4 presents two representative fitted-versus-true variance-function curves.



(a) Coefficient function under $\mathbf{A}_5(t)$.

(b) Coefficient function under $\mathbf{A}_3(t)$.

Figure 3: Representative fitted-versus-true coefficient-function curves from Case 2.



(a) $\sigma^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

(b) $\sigma^2(t) = 0.5 + 0.5 \sin^2(\pi t)$.

Figure 4: Representative fitted-versus-true variance-function curves from Case 2.

Table 5: MAE errors ($\times 10^{-2}$) for Case 2(a) - $\sigma_0^2(t) \equiv 1$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.51 ($\pm$ 0.10)	3.53 ($\pm$ 0.10)	1.55 ($\pm$ 0.65)	1.53 ($\pm$ 0.66)	18.86 ($\pm$ 2.34)	1.96 ($\pm$ 1.45)
		0.3	3.51 ($\pm$ 0.10)	3.44 ($\pm$ 0.10)	1.70 ($\pm$ 0.71)	1.69 ($\pm$ 0.71)	18.64 ($\pm$ 2.18)	6.54 ($\pm$ 2.61)
		0.6	3.51 ($\pm$ 0.10)	3.17 ($\pm$ 0.09)	1.86 ($\pm$ 0.77)	1.83 ($\pm$ 0.75)	18.61 ($\pm$ 2.62)	13.45 ($\pm$ 2.06)
		0.9	3.52 ($\pm$ 0.11)	2.58 ($\pm$ 0.08)	2.02 ($\pm$ 0.83)	1.96 ($\pm$ 0.77)	18.66 ($\pm$ 2.97)	20.39 ($\pm$ 1.34)
	5000	0.0	2.42 ($\pm$ 0.06)	2.43 ($\pm$ 0.06)	0.90 ($\pm$ 0.36)	0.91 ($\pm$ 0.37)	4.58 ($\pm$ 0.92)	1.15 ($\pm$ 0.78)
		0.3	2.39 ($\pm$ 0.06)	2.35 ($\pm$ 0.06)	0.95 ($\pm$ 0.30)	0.94 ($\pm$ 0.30)	4.61 ($\pm$ 0.98)	3.99 ($\pm$ 1.29)
		0.6	2.39 ($\pm$ 0.06)	2.20 ($\pm$ 0.06)	1.04 ($\pm$ 0.34)	1.03 ($\pm$ 0.32)	4.78 ($\pm$ 1.16)	7.73 ($\pm$ 1.04)
		0.9	2.39 ($\pm$ 0.06)	1.89 ($\pm$ 0.04)	1.14 ($\pm$ 0.37)	1.11 ($\pm$ 0.34)	5.03 ($\pm$ 1.35)	11.42 ($\pm$ 0.64)
$\mathbf{A}_2(t)$	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.52 ($\pm$ 0.72)	1.50 ($\pm$ 0.73)	19.09 ($\pm$ 2.34)	1.68 ($\pm$ 1.37)
		0.3	3.45 ($\pm$ 0.09)	3.39 ($\pm$ 0.10)	1.75 ($\pm$ 0.68)	1.73 ($\pm$ 0.67)	18.90 ($\pm$ 2.18)	6.28 ($\pm$ 2.39)
		0.6	3.45 ($\pm$ 0.09)	3.11 ($\pm$ 0.09)	1.91 ($\pm$ 0.74)	1.87 ($\pm$ 0.74)	18.89 ($\pm$ 2.55)	13.04 ($\pm$ 2.10)
		0.9	3.45 ($\pm$ 0.10)	2.47 ($\pm$ 0.08)	2.06 ($\pm$ 0.80)	1.99 ($\pm$ 0.78)	18.92 ($\pm$ 2.91)	19.93 ($\pm$ 1.51)
	5000	0.0	2.35 ($\pm$ 0.05)	2.36 ($\pm$ 0.05)	0.91 ($\pm$ 0.38)	0.91 ($\pm$ 0.39)	6.60 ($\pm$ 0.93)	1.21 ($\pm$ 0.73)
		0.3	2.33 ($\pm$ 0.05)	2.29 ($\pm$ 0.05)	0.92 ($\pm$ 0.30)	0.91 ($\pm$ 0.29)	6.61 ($\pm$ 0.96)	3.91 ($\pm$ 1.23)
		0.6	2.32 ($\pm$ 0.05)	2.12 ($\pm$ 0.04)	1.02 ($\pm$ 0.32)	1.00 ($\pm$ 0.31)	6.73 ($\pm$ 1.09)	7.63 ($\pm$ 0.99)
		0.9	2.32 ($\pm$ 0.05)	1.74 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.08 ($\pm$ 0.33)	6.88 ($\pm$ 1.27)	11.32 ($\pm$ 0.66)
$\mathbf{A}_3(t)$	2000	0.0	4.56 ($\pm$ 0.11)	4.60 ($\pm$ 0.11)	1.55 ($\pm$ 0.72)	1.55 ($\pm$ 0.71)	22.37 ($\pm$ 2.16)	1.89 ($\pm$ 1.60)
		0.3	4.56 ($\pm$ 0.11)	4.52 ($\pm$ 0.10)	1.76 ($\pm$ 0.79)	1.75 ($\pm$ 0.82)	22.36 ($\pm$ 2.18)	9.36 ($\pm$ 2.40)
		0.6	4.62 ($\pm$ 0.12)	4.25 ($\pm$ 0.10)	1.93 ($\pm$ 0.85)	1.89 ($\pm$ 0.91)	31.24 ($\pm$ 2.40)	18.48 ($\pm$ 1.71)
		0.9	4.62 ($\pm$ 0.13)	3.63 ($\pm$ 0.10)	2.06 ($\pm$ 0.90)	1.99 ($\pm$ 0.94)	31.17 ($\pm$ 2.80)	28.21 ($\pm$ 1.02)
	5000	0.0	2.70 ($\pm$ 0.05)	2.70 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.93 ($\pm$ 0.33)	13.81 ($\pm$ 1.23)	1.10 ($\pm$ 0.68)
		0.3	2.68 ($\pm$ 0.06)	2.61 ($\pm$ 0.05)	0.95 ($\pm$ 0.35)	0.94 ($\pm$ 0.35)	13.72 ($\pm$ 1.23)	4.90 ($\pm$ 1.17)
		0.6	2.68 ($\pm$ 0.06)	2.36 ($\pm$ 0.05)	1.06 ($\pm$ 0.37)	1.02 ($\pm$ 0.37)	13.78 ($\pm$ 1.33)	9.50 ($\pm$ 0.95)
		0.9	2.78 ($\pm$ 0.06)	1.68 ($\pm$ 0.04)	1.16 ($\pm$ 0.39)	1.10 ($\pm$ 0.36)	19.45 ($\pm$ 1.49)	14.72 ($\pm$ 0.59)
$\mathbf{A}_4(t)$	2000	0.0	3.13 ($\pm$ 0.09)	3.14 ($\pm$ 0.09)	1.51 ($\pm$ 0.67)	1.49 ($\pm$ 0.67)	22.56 ($\pm$ 2.39)	1.83 ($\pm$ 1.46)
		0.3	3.12 ($\pm$ 0.08)	3.03 ($\pm$ 0.08)	1.68 ($\pm$ 0.70)	1.67 ($\pm$ 0.71)	22.38 ($\pm$ 2.22)	5.44 ($\pm$ 2.37)
		0.6	3.13 ($\pm$ 0.09)	2.71 ($\pm$ 0.07)	1.85 ($\pm$ 0.77)	1.81 ($\pm$ 0.77)	22.33 ($\pm$ 2.70)	11.14 ($\pm$ 1.88)
		0.9	3.13 ($\pm$ 0.10)	1.91 ($\pm$ 0.06)	2.00 ($\pm$ 0.82)	1.94 ($\pm$ 0.80)	22.29 ($\pm$ 3.24)	16.95 ($\pm$ 1.02)
	5000	0.0	1.87 ($\pm$ 0.04)	1.87 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.91 ($\pm$ 0.34)	8.40 ($\pm$ 1.33)	1.13 ($\pm$ 0.74)
		0.3	1.84 ($\pm$ 0.05)	1.78 ($\pm$ 0.05)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.30)	8.34 ($\pm$ 1.30)	2.81 ($\pm$ 1.20)
		0.6	1.84 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.32)	8.43 ($\pm$ 1.43)	5.30 ($\pm$ 0.92)
		0.9	1.84 ($\pm$ 0.06)	1.06 ($\pm$ 0.03)	1.13 ($\pm$ 0.37)	1.10 ($\pm$ 0.34)	8.52 ($\pm$ 1.65)	7.72 ($\pm$ 0.48)
$\mathbf{A}_5(t)$	2000	0.0	5.38 ($\pm$ 0.13)	5.41 ($\pm$ 0.13)	1.69 ($\pm$ 0.65)	1.69 ($\pm$ 0.70)	22.85 ($\pm$ 1.37)	2.04 ($\pm$ 1.43)
		0.3	5.39 ($\pm$ 0.15)	5.36 ($\pm$ 0.15)	1.95 ($\pm$ 0.86)	1.88 ($\pm$ 0.93)	22.74 ($\pm$ 1.56)	11.82 ($\pm$ 2.49)
		0.6	5.39 ($\pm$ 0.16)	5.12 ($\pm$ 0.16)	2.09 ($\pm$ 0.91)	2.01 ($\pm$ 1.01)	22.77 ($\pm$ 1.78)	24.36 ($\pm$ 2.28)
		0.9	5.39 ($\pm$ 0.18)	4.66 ($\pm$ 0.17)	2.21 ($\pm$ 0.95)	2.11 ($\pm$ 1.05)	22.82 ($\pm$ 2.03)	36.95 ($\pm$ 2.04)
	5000	0.0	3.13 ($\pm$ 0.05)	3.14 ($\pm$ 0.05)	0.94 ($\pm$ 0.42)	0.95 ($\pm$ 0.43)	15.82 ($\pm$ 1.17)	1.26 ($\pm$ 0.66)
		0.3	3.11 ($\pm$ 0.05)	3.04 ($\pm$ 0.05)	1.00 ($\pm$ 0.29)	0.99 ($\pm$ 0.30)	15.76 ($\pm$ 1.25)	6.00 ($\pm$ 1.24)
		0.6	3.11 ($\pm$ 0.06)	2.77 ($\pm$ 0.04)	1.09 ($\pm$ 0.32)	1.07 ($\pm$ 0.31)	15.80 ($\pm$ 1.34)	11.97 ($\pm$ 1.03)
		0.9	3.10 ($\pm$ 0.06)	2.16 ($\pm$ 0.03)	1.19 ($\pm$ 0.35)	1.14 ($\pm$ 0.33)	15.84 ($\pm$ 1.49)	17.89 ($\pm$ 0.72)
$\mathbf{A}_6(t)$	2000	0.0	4.42 ($\pm$ 0.10)	4.44 ($\pm$ 0.11)	1.60 ($\pm$ 0.77)	1.62 ($\pm$ 0.78)	24.10 ($\pm$ 2.05)	1.86 ($\pm$ 1.08)
		0.3	4.42 ($\pm$ 0.11)	4.36 ($\pm$ 0.12)	1.76 ($\pm$ 0.73)	1.74 ($\pm$ 0.74)	24.05 ($\pm$ 2.19)	8.50 ($\pm$ 2.63)
		0.6	4.42 ($\pm$ 0.11)	4.10 ($\pm$ 0.11)	1.92 ($\pm$ 0.79)	1.88 ($\pm$ 0.80)	23.99 ($\pm$ 2.60)	17.91 ($\pm$ 2.19)
		0.9	4.55 ($\pm$ 0.13)	3.51 ($\pm$ 0.11)	2.05 ($\pm$ 0.87)	1.97 ($\pm$ 0.88)	31.82 ($\pm$ 2.90)	27.33 ($\pm$ 1.36)
	5000	0.0	2.67 ($\pm$ 0.05)	2.67 ($\pm$ 0.05)	0.93 ($\pm$ 0.35)	0.92 ($\pm$ 0.35)	14.08 ($\pm$ 1.23)	1.17 ($\pm$ 0.77)
		0.3	2.65 ($\pm$ 0.05)	2.58 ($\pm$ 0.05)	0.97 ($\pm$ 0.31)	0.95 ($\pm$ 0.32)	14.02 ($\pm$ 1.25)	4.64 ($\pm$ 1.18)
		0.6	2.65 ($\pm$ 0.05)	2.33 ($\pm$ 0.05)	1.07 ($\pm$ 0.34)	1.03 ($\pm$ 0.33)	14.07 ($\pm$ 1.36)	9.18 ($\pm$ 0.97)
		0.9	2.78 ($\pm$ 0.06)	1.67 ($\pm$ 0.04)	1.17 ($\pm$ 0.36)	1.10 ($\pm$ 0.35)	19.51 ($\pm$ 1.51)	14.59 ($\pm$ 0.55)

Table 6: MAE errors ($\times 10^{-2}$) for Case 2(b) - $\sigma_1^2(t) = 1 + 0.3 \sin(2\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.52 ($\pm$ 0.10)	3.52 ($\pm$ 0.10)	1.54 ($\pm$ 0.66)	1.52 ($\pm$ 0.66)	22.39 ($\pm$ 2.22)	1.96 ($\pm$ 1.43)
		0.3	3.51 ($\pm$ 0.09)	3.43 ($\pm$ 0.09)	1.72 ($\pm$ 0.69)	1.63 ($\pm$ 0.70)	22.33 ($\pm$ 2.07)	6.34 ($\pm$ 2.58)
		0.6	3.51 ($\pm$ 0.10)	3.13 ($\pm$ 0.09)	1.88 ($\pm$ 0.74)	1.76 ($\pm$ 0.77)	22.27 ($\pm$ 2.51)	13.21 ($\pm$ 1.97)
		0.9	3.52 ($\pm$ 0.11)	2.46 ($\pm$ 0.08)	2.02 ($\pm$ 0.80)	1.89 ($\pm$ 0.78)	22.23 ($\pm$ 2.92)	20.14 ($\pm$ 1.26)
	5000	0.0	2.28 ($\pm$ 0.06)	2.28 ($\pm$ 0.06)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.37)	7.11 ($\pm$ 1.15)	1.16 ($\pm$ 0.78)
		0.3	2.24 ($\pm$ 0.06)	2.19 ($\pm$ 0.06)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.29)	7.10 ($\pm$ 1.21)	3.62 ($\pm$ 1.23)
		0.6	2.24 ($\pm$ 0.06)	2.02 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.31)	7.20 ($\pm$ 1.32)	7.03 ($\pm$ 0.99)
		0.9	2.24 ($\pm$ 0.06)	1.64 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.10 ($\pm$ 0.33)	7.36 ($\pm$ 1.50)	10.39 ($\pm$ 0.61)
$\mathbf{A}_2(t)$	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.50 ($\pm$ 0.70)	1.47 ($\pm$ 0.70)	22.57 ($\pm$ 2.21)	1.72 ($\pm$ 1.38)
		0.3	3.46 ($\pm$ 0.10)	3.39 ($\pm$ 0.10)	1.74 ($\pm$ 0.66)	1.64 ($\pm$ 0.69)	22.53 ($\pm$ 2.07)	6.13 ($\pm$ 2.33)
		0.6	3.46 ($\pm$ 0.10)	3.09 ($\pm$ 0.09)	1.91 ($\pm$ 0.71)	1.78 ($\pm$ 0.76)	22.45 ($\pm$ 2.52)	12.87 ($\pm$ 2.01)
		0.9	3.46 ($\pm$ 0.11)	2.39 ($\pm$ 0.08)	2.06 ($\pm$ 0.77)	1.90 ($\pm$ 0.80)	22.39 ($\pm$ 2.98)	19.77 ($\pm$ 1.44)
	5000	0.0	2.22 ($\pm$ 0.05)	2.23 ($\pm$ 0.05)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.38)	6.84 ($\pm$ 1.25)	1.25 ($\pm$ 0.73)
		0.3	2.20 ($\pm$ 0.05)	2.16 ($\pm$ 0.04)	0.91 ($\pm$ 0.31)	0.90 ($\pm$ 0.29)	6.79 ($\pm$ 1.18)	3.56 ($\pm$ 1.18)
		0.6	2.19 ($\pm$ 0.05)	1.97 ($\pm$ 0.04)	1.01 ($\pm$ 0.34)	0.99 ($\pm$ 0.31)	6.90 ($\pm$ 1.29)	6.97 ($\pm$ 0.94)
		0.9	2.19 ($\pm$ 0.05)	1.54 ($\pm$ 0.03)	1.11 ($\pm$ 0.36)	1.07 ($\pm$ 0.32)	7.05 ($\pm$ 1.46)	10.34 ($\pm$ 0.62)
$\mathbf{A}_3(t)$	2000	0.0	4.56 ($\pm$ 0.11)	4.58 ($\pm$ 0.11)	1.55 ($\pm$ 0.70)	1.50 ($\pm$ 0.71)	22.24 ($\pm$ 2.16)	1.97 ($\pm$ 1.52)
		0.3	4.56 ($\pm$ 0.10)	4.51 ($\pm$ 0.10)	1.77 ($\pm$ 0.77)	1.72 ($\pm$ 0.82)	22.39 ($\pm$ 2.12)	9.33 ($\pm$ 2.34)
		0.6	4.61 ($\pm$ 0.12)	4.24 ($\pm$ 0.10)	1.94 ($\pm$ 0.83)	1.85 ($\pm$ 0.89)	31.19 ($\pm$ 2.41)	18.46 ($\pm$ 1.65)
		0.9	4.62 ($\pm$ 0.13)	3.62 ($\pm$ 0.09)	2.07 ($\pm$ 0.87)	1.94 ($\pm$ 0.92)	31.08 ($\pm$ 2.79)	28.22 ($\pm$ 1.05)
	5000	0.0	2.69 ($\pm$ 0.05)	2.69 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.92 ($\pm$ 0.34)	13.72 ($\pm$ 1.22)	1.15 ($\pm$ 0.76)
		0.3	2.68 ($\pm$ 0.06)	2.61 ($\pm$ 0.05)	0.95 ($\pm$ 0.35)	0.93 ($\pm$ 0.34)	13.66 ($\pm$ 1.25)	4.84 ($\pm$ 1.14)
		0.6	2.68 ($\pm$ 0.06)	2.36 ($\pm$ 0.05)	1.05 ($\pm$ 0.38)	1.01 ($\pm$ 0.35)	13.71 ($\pm$ 1.34)	9.45 ($\pm$ 0.94)
		0.9	2.77 ($\pm$ 0.06)	1.67 ($\pm$ 0.04)	1.16 ($\pm$ 0.39)	1.08 ($\pm$ 0.35)	19.35 ($\pm$ 1.49)	14.69 ($\pm$ 0.59)
$\mathbf{A}_4(t)$	2000	0.0	3.25 ($\pm$ 0.09)	3.26 ($\pm$ 0.09)	1.49 ($\pm$ 0.68)	1.48 ($\pm$ 0.67)	24.79 ($\pm$ 2.24)	1.84 ($\pm$ 1.49)
		0.3	3.25 ($\pm$ 0.08)	3.15 ($\pm$ 0.08)	1.70 ($\pm$ 0.69)	1.61 ($\pm$ 0.72)	24.76 ($\pm$ 2.11)	5.60 ($\pm$ 2.34)
		0.6	3.25 ($\pm$ 0.09)	2.82 ($\pm$ 0.07)	1.86 ($\pm$ 0.75)	1.75 ($\pm$ 0.78)	24.68 ($\pm$ 2.57)	11.64 ($\pm$ 1.82)
		0.9	3.26 ($\pm$ 0.10)	2.00 ($\pm$ 0.06)	2.01 ($\pm$ 0.80)	1.88 ($\pm$ 0.81)	24.61 ($\pm$ 3.06)	17.83 ($\pm$ 1.03)
	5000	0.0	1.89 ($\pm$ 0.05)	1.89 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.90 ($\pm$ 0.34)	9.48 ($\pm$ 1.31)	1.20 ($\pm$ 0.72)
		0.3	1.86 ($\pm$ 0.05)	1.80 ($\pm$ 0.05)	0.92 ($\pm$ 0.31)	0.91 ($\pm$ 0.29)	9.44 ($\pm$ 1.26)	2.82 ($\pm$ 1.16)
		0.6	1.86 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.02 ($\pm$ 0.34)	1.00 ($\pm$ 0.31)	9.53 ($\pm$ 1.37)	5.38 ($\pm$ 0.90)
		0.9	1.86 ($\pm$ 0.06)	1.02 ($\pm$ 0.03)	1.11 ($\pm$ 0.37)	1.08 ($\pm$ 0.32)	9.64 ($\pm$ 1.55)	7.89 ($\pm$ 0.50)
$\mathbf{A}_5(t)$	2000	0.0	5.34 ($\pm$ 0.12)	5.38 ($\pm$ 0.13)	1.69 ($\pm$ 0.63)	1.64 ($\pm$ 0.68)	20.41 ($\pm$ 1.81)	1.99 ($\pm$ 1.48)
		0.3	5.36 ($\pm$ 0.15)	5.32 ($\pm$ 0.15)	1.96 ($\pm$ 0.82)	1.81 ($\pm$ 0.90)	20.48 ($\pm$ 2.03)	11.74 ($\pm$ 2.52)
		0.6	5.36 ($\pm$ 0.16)	5.09 ($\pm$ 0.16)	2.10 ($\pm$ 0.87)	1.94 ($\pm$ 0.98)	20.44 ($\pm$ 2.30)	24.30 ($\pm$ 2.31)
		0.9	5.36 ($\pm$ 0.18)	4.63 ($\pm$ 0.16)	2.22 ($\pm$ 0.91)	2.03 ($\pm$ 1.03)	20.40 ($\pm$ 2.57)	36.92 ($\pm$ 2.09)
	5000	0.0	3.11 ($\pm$ 0.05)	3.11 ($\pm$ 0.05)	0.94 ($\pm$ 0.40)	0.93 ($\pm$ 0.43)	15.72 ($\pm$ 1.16)	1.31 ($\pm$ 0.71)
		0.3	3.09 ($\pm$ 0.05)	3.02 ($\pm$ 0.05)	0.98 ($\pm$ 0.30)	0.97 ($\pm$ 0.31)	15.70 ($\pm$ 1.29)	5.98 ($\pm$ 1.21)
		0.6	3.08 ($\pm$ 0.05)	2.75 ($\pm$ 0.04)	1.08 ($\pm$ 0.33)	1.05 ($\pm$ 0.32)	15.72 ($\pm$ 1.37)	11.95 ($\pm$ 1.01)
		0.9	3.08 ($\pm$ 0.06)	2.15 ($\pm$ 0.03)	1.17 ($\pm$ 0.36)	1.12 ($\pm$ 0.33)	15.75 ($\pm$ 1.50)	17.89 ($\pm$ 0.70)
$\mathbf{A}_6(t)$	2000	0.0	4.40 ($\pm$ 0.10)	4.42 ($\pm$ 0.10)	1.61 ($\pm$ 0.74)	1.61 ($\pm$ 0.76)	23.98 ($\pm$ 2.01)	1.83 ($\pm$ 1.08)
		0.3	4.40 ($\pm$ 0.11)	4.35 ($\pm$ 0.12)	1.79 ($\pm$ 0.70)	1.68 ($\pm$ 0.72)	24.06 ($\pm$ 2.15)	8.40 ($\pm$ 2.57)
		0.6	4.40 ($\pm$ 0.11)	4.08 ($\pm$ 0.11)	1.94 ($\pm$ 0.76)	1.81 ($\pm$ 0.79)	23.96 ($\pm$ 2.56)	17.82 ($\pm$ 2.12)
		0.9	4.53 ($\pm$ 0.13)	3.50 ($\pm$ 0.11)	2.08 ($\pm$ 0.84)	1.90 ($\pm$ 0.87)	31.74 ($\pm$ 2.88)	27.29 ($\pm$ 1.34)
	5000	0.0	2.66 ($\pm$ 0.05)	2.66 ($\pm$ 0.05)	0.93 ($\pm$ 0.34)	0.91 ($\pm$ 0.36)	13.99 ($\pm$ 1.23)	1.23 ($\pm$ 0.78)
		0.3	2.64 ($\pm$ 0.05)	2.57 ($\pm$ 0.05)	0.96 ($\pm$ 0.31)	0.94 ($\pm$ 0.31)	13.96 ($\pm$ 1.28)	4.60 ($\pm$ 1.14)
		0.6	2.64 ($\pm$ 0.05)	2.32 ($\pm$ 0.05)	1.06 ($\pm$ 0.34)	1.02 ($\pm$ 0.32)	14.00 ($\pm$ 1.38)	9.15 ($\pm$ 0.92)
		0.9	2.76 ($\pm$ 0.06)	1.66 ($\pm$ 0.04)	1.16 ($\pm$ 0.37)	1.09 ($\pm$ 0.34)	19.41 ($\pm$ 1.51)	14.58 ($\pm$ 0.51)

Table 7: MAE errors ($\times 10^{-2}$) for Case 2(c) - $\sigma_2^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.60 ($\pm$ 0.09)	3.60 ($\pm$ 0.09)	1.51 ($\pm$ 0.68)	1.52 ($\pm$ 0.68)	23.71 ($\pm$ 2.11)	1.91 ($\pm$ 1.38)
		0.3	3.59 ($\pm$ 0.10)	3.50 ($\pm$ 0.10)	1.70 ($\pm$ 0.71)	1.67 ($\pm$ 0.70)	23.57 ($\pm$ 1.96)	6.71 ($\pm$ 2.51)
		0.6	3.59 ($\pm$ 0.10)	3.20 ($\pm$ 0.09)	1.87 ($\pm$ 0.77)	1.81 ($\pm$ 0.76)	23.57 ($\pm$ 2.36)	13.85 ($\pm$ 1.93)
		0.9	3.59 ($\pm$ 0.11)	2.53 ($\pm$ 0.08)	2.02 ($\pm$ 0.82)	1.93 ($\pm$ 0.78)	23.61 ($\pm$ 2.78)	21.00 ($\pm$ 1.23)
	5000	0.0	2.32 ($\pm$ 0.06)	2.31 ($\pm$ 0.06)	0.92 ($\pm$ 0.37)	0.89 ($\pm$ 0.36)	8.74 ($\pm$ 1.05)	1.06 ($\pm$ 0.74)
		0.3	2.29 ($\pm$ 0.06)	2.23 ($\pm$ 0.06)	0.94 ($\pm$ 0.30)	0.93 ($\pm$ 0.29)	8.81 ($\pm$ 1.05)	3.82 ($\pm$ 1.32)
		0.6	2.28 ($\pm$ 0.06)	2.05 ($\pm$ 0.05)	1.04 ($\pm$ 0.33)	1.02 ($\pm$ 0.31)	8.95 ($\pm$ 1.24)	7.32 ($\pm$ 1.06)
		0.9	2.28 ($\pm$ 0.06)	1.66 ($\pm$ 0.04)	1.14 ($\pm$ 0.37)	1.10 ($\pm$ 0.33)	9.11 ($\pm$ 1.49)	10.76 ($\pm$ 0.63)
$\mathbf{A}_2(t)$	2000	0.0	3.56 ($\pm$ 0.10)	3.57 ($\pm$ 0.10)	1.48 ($\pm$ 0.73)	1.49 ($\pm$ 0.72)	23.95 ($\pm$ 2.07)	1.66 ($\pm$ 1.32)
		0.3	3.56 ($\pm$ 0.10)	3.47 ($\pm$ 0.10)	1.73 ($\pm$ 0.69)	1.70 ($\pm$ 0.69)	23.82 ($\pm$ 1.90)	6.48 ($\pm$ 2.41)
		0.6	3.56 ($\pm$ 0.10)	3.17 ($\pm$ 0.09)	1.89 ($\pm$ 0.75)	1.83 ($\pm$ 0.75)	23.85 ($\pm$ 2.25)	13.54 ($\pm$ 1.96)
		0.9	3.56 ($\pm$ 0.11)	2.46 ($\pm$ 0.08)	2.05 ($\pm$ 0.80)	1.95 ($\pm$ 0.80)	23.91 ($\pm$ 2.65)	20.66 ($\pm$ 1.36)
	5000	0.0	2.28 ($\pm$ 0.06)	2.27 ($\pm$ 0.05)	0.92 ($\pm$ 0.38)	0.90 ($\pm$ 0.37)	10.41 ($\pm$ 1.04)	1.15 ($\pm$ 0.71)
		0.3	2.26 ($\pm$ 0.05)	2.20 ($\pm$ 0.04)	0.92 ($\pm$ 0.31)	0.92 ($\pm$ 0.28)	10.45 ($\pm$ 1.07)	3.76 ($\pm$ 1.24)
		0.6	2.26 ($\pm$ 0.05)	2.00 ($\pm$ 0.04)	1.02 ($\pm$ 0.34)	1.00 ($\pm$ 0.31)	10.56 ($\pm$ 1.24)	7.25 ($\pm$ 0.99)
		0.9	2.25 ($\pm$ 0.05)	1.55 ($\pm$ 0.03)	1.12 ($\pm$ 0.36)	1.08 ($\pm$ 0.32)	10.69 ($\pm$ 1.48)	10.69 ($\pm$ 0.62)
$\mathbf{A}_3(t)$	2000	0.0	4.63 ($\pm$ 0.11)	4.66 ($\pm$ 0.11)	1.54 ($\pm$ 0.71)	1.52 ($\pm$ 0.71)	23.47 ($\pm$ 2.06)	1.91 ($\pm$ 1.56)
		0.3	4.63 ($\pm$ 0.11)	4.59 ($\pm$ 0.10)	1.75 ($\pm$ 0.81)	1.73 ($\pm$ 0.80)	23.53 ($\pm$ 2.03)	9.63 ($\pm$ 2.37)
		0.6	4.69 ($\pm$ 0.13)	4.32 ($\pm$ 0.11)	1.93 ($\pm$ 0.87)	1.86 ($\pm$ 0.88)	32.34 ($\pm$ 2.26)	18.91 ($\pm$ 1.69)
		0.9	4.69 ($\pm$ 0.14)	3.70 ($\pm$ 0.10)	2.07 ($\pm$ 0.91)	1.95 ($\pm$ 0.91)	32.29 ($\pm$ 2.68)	28.85 ($\pm$ 1.03)
	5000	0.0	2.72 ($\pm$ 0.05)	2.72 ($\pm$ 0.05)	0.94 ($\pm$ 0.33)	0.92 ($\pm$ 0.31)	14.38 ($\pm$ 1.22)	1.07 ($\pm$ 0.70)
		0.3	2.70 ($\pm$ 0.06)	2.63 ($\pm$ 0.05)	0.95 ($\pm$ 0.36)	0.94 ($\pm$ 0.34)	14.32 ($\pm$ 1.21)	5.05 ($\pm$ 1.21)
		0.6	2.70 ($\pm$ 0.06)	2.38 ($\pm$ 0.05)	1.05 ($\pm$ 0.38)	1.02 ($\pm$ 0.36)	14.40 ($\pm$ 1.34)	9.74 ($\pm$ 0.98)
		0.9	2.80 ($\pm$ 0.06)	1.69 ($\pm$ 0.04)	1.16 ($\pm$ 0.40)	1.09 ($\pm$ 0.35)	20.05 ($\pm$ 1.52)	14.99 ($\pm$ 0.61)
$\mathbf{A}_4(t)$	2000	0.0	3.33 ($\pm$ 0.09)	3.33 ($\pm$ 0.09)	1.48 ($\pm$ 0.70)	1.49 ($\pm$ 0.69)	26.11 ($\pm$ 2.14)	1.81 ($\pm$ 1.44)
		0.3	3.32 ($\pm$ 0.09)	3.23 ($\pm$ 0.08)	1.69 ($\pm$ 0.71)	1.65 ($\pm$ 0.71)	25.99 ($\pm$ 2.02)	5.93 ($\pm$ 2.42)
		0.6	3.32 ($\pm$ 0.09)	2.89 ($\pm$ 0.08)	1.86 ($\pm$ 0.77)	1.79 ($\pm$ 0.78)	25.97 ($\pm$ 2.46)	12.28 ($\pm$ 1.82)
		0.9	3.33 ($\pm$ 0.11)	2.06 ($\pm$ 0.07)	2.01 ($\pm$ 0.82)	1.91 ($\pm$ 0.80)	25.98 ($\pm$ 2.93)	18.68 ($\pm$ 1.01)
	5000	0.0	1.93 ($\pm$ 0.04)	1.93 ($\pm$ 0.04)	0.92 ($\pm$ 0.34)	0.89 ($\pm$ 0.33)	10.46 ($\pm$ 1.17)	1.09 ($\pm$ 0.70)
		0.3	1.90 ($\pm$ 0.05)	1.83 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.92 ($\pm$ 0.29)	10.47 ($\pm$ 1.14)	3.03 ($\pm$ 1.25)
		0.6	1.90 ($\pm$ 0.06)	1.60 ($\pm$ 0.05)	1.03 ($\pm$ 0.34)	1.01 ($\pm$ 0.31)	10.59 ($\pm$ 1.30)	5.68 ($\pm$ 0.96)
		0.9	1.90 ($\pm$ 0.06)	1.04 ($\pm$ 0.03)	1.13 ($\pm$ 0.37)	1.09 ($\pm$ 0.33)	10.72 ($\pm$ 1.54)	8.25 ($\pm$ 0.51)
$\mathbf{A}_5(t)$	2000	0.0	5.45 ($\pm$ 0.13)	5.49 ($\pm$ 0.13)	1.68 ($\pm$ 0.66)	1.67 ($\pm$ 0.70)	26.59 ($\pm$ 1.30)	2.02 ($\pm$ 1.32)
		0.3	5.47 ($\pm$ 0.16)	5.44 ($\pm$ 0.16)	1.95 ($\pm$ 0.88)	1.86 ($\pm$ 0.92)	26.57 ($\pm$ 1.45)	12.05 ($\pm$ 2.44)
		0.6	5.47 ($\pm$ 0.17)	5.20 ($\pm$ 0.16)	2.09 ($\pm$ 0.94)	1.98 ($\pm$ 1.00)	26.61 ($\pm$ 1.65)	24.82 ($\pm$ 2.24)
		0.9	5.48 ($\pm$ 0.18)	4.74 ($\pm$ 0.17)	2.22 ($\pm$ 0.97)	2.09 ($\pm$ 1.04)	26.69 ($\pm$ 1.91)	37.63 ($\pm$ 2.03)
	5000	0.0	3.16 ($\pm$ 0.05)	3.16 ($\pm$ 0.05)	0.95 ($\pm$ 0.43)	0.94 ($\pm$ 0.41)	16.81 ($\pm$ 1.06)	1.23 ($\pm$ 0.66)
		0.3	3.14 ($\pm$ 0.05)	3.06 ($\pm$ 0.05)	0.99 ($\pm$ 0.31)	0.98 ($\pm$ 0.28)	16.77 ($\pm$ 1.09)	6.14 ($\pm$ 1.25)
		0.6	3.13 ($\pm$ 0.06)	2.80 ($\pm$ 0.04)	1.09 ($\pm$ 0.34)	1.06 ($\pm$ 0.30)	16.83 ($\pm$ 1.21)	12.18 ($\pm$ 1.05)
		0.9	3.13 ($\pm$ 0.06)	2.18 ($\pm$ 0.03)	1.18 ($\pm$ 0.36)	1.13 ($\pm$ 0.32)	16.90 ($\pm$ 1.38)	18.17 ($\pm$ 0.74)
$\mathbf{A}_6(t)$	2000	0.0	4.46 ($\pm$ 0.10)	4.48 ($\pm$ 0.11)	1.58 ($\pm$ 0.78)	1.59 ($\pm$ 0.76)	25.20 ($\pm$ 1.96)	1.81 ($\pm$ 1.11)
		0.3	4.46 ($\pm$ 0.11)	4.41 ($\pm$ 0.12)	1.76 ($\pm$ 0.73)	1.72 ($\pm$ 0.73)	25.20 ($\pm$ 2.05)	8.76 ($\pm$ 2.61)
		0.6	4.46 ($\pm$ 0.11)	4.14 ($\pm$ 0.11)	1.92 ($\pm$ 0.79)	1.85 ($\pm$ 0.80)	25.16 ($\pm$ 2.46)	18.40 ($\pm$ 2.15)
		0.9	4.60 ($\pm$ 0.13)	3.57 ($\pm$ 0.12)	2.05 ($\pm$ 0.88)	1.94 ($\pm$ 0.87)	32.94 ($\pm$ 2.77)	27.95 ($\pm$ 1.33)
	5000	0.0	2.68 ($\pm$ 0.05)	2.68 ($\pm$ 0.05)	0.93 ($\pm$ 0.36)	0.90 ($\pm$ 0.34)	14.65 ($\pm$ 1.22)	1.15 ($\pm$ 0.76)
		0.3	2.66 ($\pm$ 0.05)	2.59 ($\pm$ 0.05)	0.96 ($\pm$ 0.32)	0.94 ($\pm$ 0.31)	14.62 ($\pm$ 1.25)	4.79 ($\pm$ 1.21)
		0.6	2.65 ($\pm$ 0.05)	2.33 ($\pm$ 0.05)	1.06 ($\pm$ 0.35)	1.02 ($\pm$ 0.32)	14.69 ($\pm$ 1.37)	9.41 ($\pm$ 0.98)
		0.9	2.79 ($\pm$ 0.06)	1.67 ($\pm$ 0.04)	1.17 ($\pm$ 0.38)	1.09 ($\pm$ 0.34)	20.11 ($\pm$ 1.53)	14.86 ($\pm$ 0.55)

Table 8: MAE errors ($\times 10^{-2}$) for Case 2(d) - $\sigma_3^2(t) = 0.5 + 0.5 \sin^2(\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.03 ($\pm$ 0.09)	3.04 ($\pm$ 0.09)	1.38 ($\pm$ 0.59)	1.29 ($\pm$ 0.55)	15.01 ($\pm$ 1.82)	2.13 ($\pm$ 1.48)
		0.3	3.02 ($\pm$ 0.08)	2.96 ($\pm$ 0.08)	1.48 ($\pm$ 0.61)	1.41 ($\pm$ 0.60)	14.82 ($\pm$ 1.76)	6.41 ($\pm$ 2.66)
		0.6	3.03 ($\pm$ 0.09)	2.72 ($\pm$ 0.08)	1.61 ($\pm$ 0.68)	1.53 ($\pm$ 0.64)	14.76 ($\pm$ 2.15)	13.17 ($\pm$ 2.18)
		0.9	3.03 ($\pm$ 0.09)	2.19 ($\pm$ 0.07)	1.74 ($\pm$ 0.74)	1.64 ($\pm$ 0.65)	14.73 ($\pm$ 2.57)	20.00 ($\pm$ 1.39)
	5000	0.0	2.03 ($\pm$ 0.05)	2.04 ($\pm$ 0.05)	0.78 ($\pm$ 0.30)	0.77 ($\pm$ 0.32)	3.64 ($\pm$ 0.97)	1.21 ($\pm$ 0.88)
		0.3	2.00 ($\pm$ 0.05)	1.98 ($\pm$ 0.05)	0.82 ($\pm$ 0.26)	0.79 ($\pm$ 0.25)	3.57 ($\pm$ 0.92)	3.76 ($\pm$ 1.33)
		0.6	2.00 ($\pm$ 0.05)	1.84 ($\pm$ 0.05)	0.91 ($\pm$ 0.29)	0.86 ($\pm$ 0.27)	3.67 ($\pm$ 0.98)	7.37 ($\pm$ 1.07)
		0.9	2.00 ($\pm$ 0.05)	1.57 ($\pm$ 0.03)	0.99 ($\pm$ 0.31)	0.93 ($\pm$ 0.29)	3.81 ($\pm$ 1.07)	10.93 ($\pm$ 0.66)
$\mathbf{A}_2(t)$	2000	0.0	2.98 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.33 ($\pm$ 0.62)	1.24 ($\pm$ 0.61)	15.17 ($\pm$ 1.84)	1.86 ($\pm$ 1.54)
		0.3	2.98 ($\pm$ 0.08)	2.92 ($\pm$ 0.09)	1.50 ($\pm$ 0.59)	1.43 ($\pm$ 0.59)	15.01 ($\pm$ 1.79)	6.22 ($\pm$ 2.33)
		0.6	2.98 ($\pm$ 0.09)	2.68 ($\pm$ 0.08)	1.64 ($\pm$ 0.64)	1.54 ($\pm$ 0.65)	14.96 ($\pm$ 2.16)	12.81 ($\pm$ 2.16)
		0.9	2.98 ($\pm$ 0.09)	2.11 ($\pm$ 0.07)	1.78 ($\pm$ 0.70)	1.65 ($\pm$ 0.66)	14.92 ($\pm$ 2.56)	19.60 ($\pm$ 1.53)
	5000	0.0	1.98 ($\pm$ 0.04)	1.99 ($\pm$ 0.04)	0.78 ($\pm$ 0.31)	0.77 ($\pm$ 0.33)	4.30 ($\pm$ 0.79)	1.32 ($\pm$ 0.69)
		0.3	1.96 ($\pm$ 0.04)	1.93 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.76 ($\pm$ 0.26)	4.23 ($\pm$ 0.81)	3.70 ($\pm$ 1.30)
		0.6	1.95 ($\pm$ 0.04)	1.78 ($\pm$ 0.04)	0.90 ($\pm$ 0.27)	0.83 ($\pm$ 0.28)	4.30 ($\pm$ 0.88)	7.30 ($\pm$ 1.05)
		0.9	1.95 ($\pm$ 0.05)	1.44 ($\pm$ 0.03)	0.98 ($\pm$ 0.30)	0.89 ($\pm$ 0.29)	4.41 ($\pm$ 0.98)	10.86 ($\pm$ 0.69)
$\mathbf{A}_3(t)$	2000	0.0	4.00 ($\pm$ 0.10)	4.01 ($\pm$ 0.10)	1.37 ($\pm$ 0.61)	1.34 ($\pm$ 0.58)	21.36 ($\pm$ 1.46)	2.23 ($\pm$ 1.52)
		0.3	4.00 ($\pm$ 0.09)	3.94 ($\pm$ 0.09)	1.55 ($\pm$ 0.65)	1.50 ($\pm$ 0.71)	21.30 ($\pm$ 1.65)	9.07 ($\pm$ 2.43)
		0.6	4.00 ($\pm$ 0.10)	3.70 ($\pm$ 0.08)	1.68 ($\pm$ 0.71)	1.60 ($\pm$ 0.78)	21.23 ($\pm$ 1.96)	18.91 ($\pm$ 1.81)
		0.9	4.00 ($\pm$ 0.11)	3.21 ($\pm$ 0.08)	1.79 ($\pm$ 0.77)	1.67 ($\pm$ 0.80)	21.17 ($\pm$ 2.27)	28.79 ($\pm$ 1.11)
	5000	0.0	2.38 ($\pm$ 0.05)	2.38 ($\pm$ 0.05)	0.80 ($\pm$ 0.25)	0.79 ($\pm$ 0.29)	9.03 ($\pm$ 0.96)	1.14 ($\pm$ 0.74)
		0.3	2.37 ($\pm$ 0.05)	2.31 ($\pm$ 0.05)	0.83 ($\pm$ 0.29)	0.79 ($\pm$ 0.31)	8.95 ($\pm$ 0.96)	5.00 ($\pm$ 1.20)
		0.6	2.40 ($\pm$ 0.05)	2.08 ($\pm$ 0.04)	0.92 ($\pm$ 0.31)	0.86 ($\pm$ 0.32)	13.69 ($\pm$ 1.03)	9.88 ($\pm$ 0.93)
		0.9	2.40 ($\pm$ 0.05)	1.48 ($\pm$ 0.03)	1.00 ($\pm$ 0.32)	0.91 ($\pm$ 0.31)	13.71 ($\pm$ 1.14)	14.73 ($\pm$ 0.59)
$\mathbf{A}_4(t)$	2000	0.0	2.75 ($\pm$ 0.08)	2.75 ($\pm$ 0.08)	1.34 ($\pm$ 0.60)	1.24 ($\pm$ 0.56)	17.41 ($\pm$ 1.85)	1.98 ($\pm$ 1.59)
		0.3	2.74 ($\pm$ 0.07)	2.66 ($\pm$ 0.07)	1.46 ($\pm$ 0.61)	1.40 ($\pm$ 0.62)	17.24 ($\pm$ 1.81)	5.47 ($\pm$ 2.38)
		0.6	2.75 ($\pm$ 0.08)	2.38 ($\pm$ 0.06)	1.60 ($\pm$ 0.68)	1.52 ($\pm$ 0.66)	17.19 ($\pm$ 2.18)	11.14 ($\pm$ 2.00)
		0.9	2.75 ($\pm$ 0.09)	1.69 ($\pm$ 0.05)	1.74 ($\pm$ 0.73)	1.62 ($\pm$ 0.67)	17.15 ($\pm$ 2.60)	16.99 ($\pm$ 1.07)
	5000	0.0	1.62 ($\pm$ 0.04)	1.62 ($\pm$ 0.04)	0.78 ($\pm$ 0.28)	0.77 ($\pm$ 0.30)	6.40 ($\pm$ 1.03)	1.20 ($\pm$ 0.78)
		0.3	1.60 ($\pm$ 0.05)	1.55 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.77 ($\pm$ 0.26)	6.33 ($\pm$ 0.98)	2.74 ($\pm$ 1.19)
		0.6	1.60 ($\pm$ 0.05)	1.36 ($\pm$ 0.04)	0.90 ($\pm$ 0.28)	0.85 ($\pm$ 0.28)	6.39 ($\pm$ 1.06)	5.22 ($\pm$ 0.96)
		0.9	1.60 ($\pm$ 0.05)	0.91 ($\pm$ 0.03)	0.98 ($\pm$ 0.31)	0.92 ($\pm$ 0.29)	6.46 ($\pm$ 1.21)	7.66 ($\pm$ 0.51)
$\mathbf{A}_5(t)$	2000	0.0	4.80 ($\pm$ 0.11)	4.84 ($\pm$ 0.11)	1.49 ($\pm$ 0.58)	1.44 ($\pm$ 0.58)	15.36 ($\pm$ 1.29)	2.16 ($\pm$ 1.55)
		0.3	4.81 ($\pm$ 0.13)	4.79 ($\pm$ 0.13)	1.71 ($\pm$ 0.73)	1.59 ($\pm$ 0.81)	15.14 ($\pm$ 1.36)	12.72 ($\pm$ 2.63)
		0.6	4.81 ($\pm$ 0.14)	4.61 ($\pm$ 0.14)	1.83 ($\pm$ 0.77)	1.71 ($\pm$ 0.87)	15.17 ($\pm$ 1.47)	26.05 ($\pm$ 2.49)
		0.9	4.81 ($\pm$ 0.15)	4.26 ($\pm$ 0.15)	1.93 ($\pm$ 0.81)	1.80 ($\pm$ 0.91)	15.22 ($\pm$ 1.61)	39.43 ($\pm$ 2.29)
	5000	0.0	2.77 ($\pm$ 0.05)	2.77 ($\pm$ 0.05)	0.82 ($\pm$ 0.33)	0.80 ($\pm$ 0.38)	10.16 ($\pm$ 0.87)	1.34 ($\pm$ 0.74)
		0.3	2.75 ($\pm$ 0.05)	2.69 ($\pm$ 0.04)	0.87 ($\pm$ 0.24)	0.83 ($\pm$ 0.26)	10.10 ($\pm$ 0.96)	6.26 ($\pm$ 1.32)
		0.6	2.74 ($\pm$ 0.05)	2.48 ($\pm$ 0.04)	0.96 ($\pm$ 0.27)	0.90 ($\pm$ 0.27)	10.12 ($\pm$ 1.02)	12.57 ($\pm$ 1.10)
		0.9	2.74 ($\pm$ 0.05)	1.99 ($\pm$ 0.03)	1.04 ($\pm$ 0.29)	0.96 ($\pm$ 0.28)	10.15 ($\pm$ 1.13)	18.86 ($\pm$ 0.77)
$\mathbf{A}_6(t)$	2000	0.0	3.93 ($\pm$ 0.09)	3.94 ($\pm$ 0.09)	1.41 ($\pm$ 0.67)	1.40 ($\pm$ 0.68)	15.22 ($\pm$ 1.60)	1.94 ($\pm$ 1.20)
		0.3	3.93 ($\pm$ 0.10)	3.88 ($\pm$ 0.10)	1.54 ($\pm$ 0.63)	1.48 ($\pm$ 0.61)	15.11 ($\pm$ 1.83)	8.87 ($\pm$ 2.71)
		0.6	3.96 ($\pm$ 0.10)	3.65 ($\pm$ 0.10)	1.65 ($\pm$ 0.72)	1.57 ($\pm$ 0.70)	21.89 ($\pm$ 2.07)	18.05 ($\pm$ 2.10)
		0.9	3.95 ($\pm$ 0.11)	3.11 ($\pm$ 0.09)	1.77 ($\pm$ 0.77)	1.65 ($\pm$ 0.73)	21.83 ($\pm$ 2.36)	27.72 ($\pm$ 1.50)
	5000	0.0	2.37 ($\pm$ 0.04)	2.36 ($\pm$ 0.04)	0.79 ($\pm$ 0.29)	0.79 ($\pm$ 0.30)	9.31 ($\pm$ 0.96)	1.27 ($\pm$ 0.78)
		0.3	2.35 ($\pm$ 0.05)	2.29 ($\pm$ 0.04)	0.84 ($\pm$ 0.26)	0.80 ($\pm$ 0.28)	9.26 ($\pm$ 0.96)	4.70 ($\pm$ 1.24)
		0.6	2.35 ($\pm$ 0.05)	2.09 ($\pm$ 0.04)	0.93 ($\pm$ 0.28)	0.86 ($\pm$ 0.29)	9.29 ($\pm$ 1.02)	9.40 ($\pm$ 1.02)
		0.9	2.40 ($\pm$ 0.05)	1.49 ($\pm$ 0.03)	1.01 ($\pm$ 0.30)	0.92 ($\pm$ 0.30)	13.77 ($\pm$ 1.15)	14.57 ($\pm$ 0.58)

EMPIRICAL EXPERIMENTS

Dataset Information. The glaucoma real-world appraisal progression ensemble (GRAPE) is a longitudinal multimodal dataset collected for glaucoma management. It contains fundus images together with clinical information collected during follow-up. For each image visit, two image types data are collected: a color fundus photograph (CFP), an optic-disc-centered region-of-interest (ROI) image. An example is given in Figure 5.

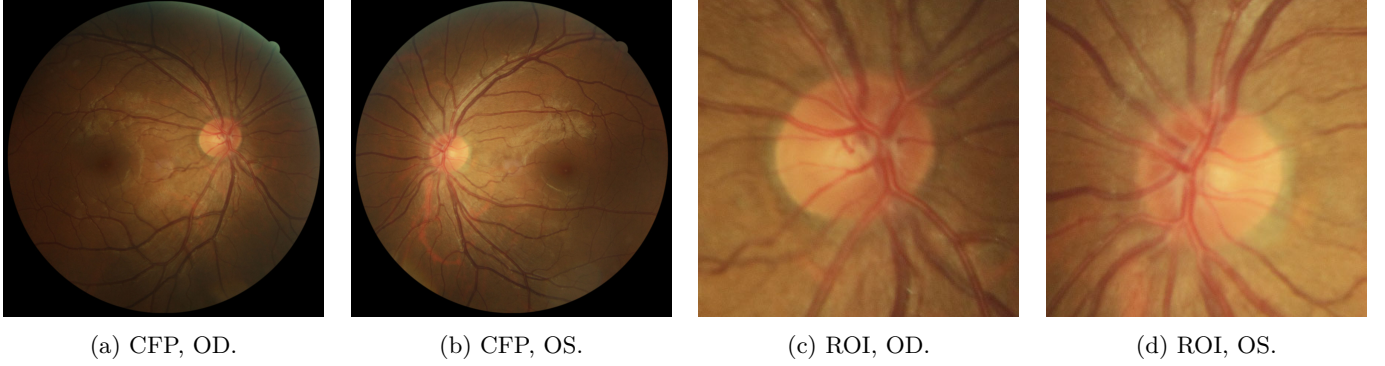


Figure 5: Original CFP and ROI images from an representative paired visit.

The intraocular pressure (IOP), age, gender, and visual-field (VF) measurements are also included in the dataset. Since elevated IOP is an important biomarker and modifiable risk factor for glaucoma progression, our objective is to quantify how the association between fundus images and IOP varies with age and to evaluate the predictive value of the image and clinical covariates.

Preprocessing. The distributed image subset contains 631 eye-level visits with matched CFP and ROI images. We paired oculus dexter (OD) and oculus sinister (OS) records from the same subject at the same follow-up time, obtaining 276 paired visits. After excluding a paired visit whenever the IOP of either eye exceeded 35 mmHg, the primary analysis retained 273 paired visits, corresponding to 546 eye-level observations for each image representation. All images were resized to $192 \times 192 \times 3$, and OS images were horizontally flipped to align with the OD images.

Let $i = 1, \dots, 273$ index paired visits and $j \in \{1, 2\}$ index the OD and OS eyes, respectively. For either the CFP or ROI analysis, let $\mathcal{X}_{ij} \in \mathbb{R}^{192 \times 192 \times 3}$ denote the image tensor and let y_{ij} denote the corresponding IOP. The index variable t_i is age, min-max normalized to $[0, 1]$. The pair-level covariate vector $\mathbf{z}_i \in \mathbb{R}^{60}$ contains gender and the paired-eye averages of 59 non-blind-spot VF measurements; the continuous VF covariates and IOP outcomes are standardized before model fitting. We then apply the paired-eye VCTR model

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}, \quad \text{where} \quad \begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \sigma^2(t_i) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right),$$

and $\mathcal{A}(t)$ captures the age-dependent image effect and $\boldsymbol{\beta}$ contains the constant effects of the clinical covariates.

Choosing Hyperparameter. We select four hyperparameters separately for CFP and ROI: the partition grid $S^{(1)} \times S^{(2)} \times S^{(3)}$, the blockwise CP rank R , the bandwidth h for estimating the varying image effect, and the bandwidth $\bar{h}$ for estimating the variance function. The candidate feature set contains 17 partition grids and 56 available (S, R) combinations. For each feature configuration, we consider 32 values of h ranging from 0.18 to 8.00 and 15 values of $\bar{h}$ ranging from 0.20 to 4.00. This gives $56 \times 32 \times 15 = 26,880$ candidates for each image type and 53,760 candidates in total. Hyperparameters are selected using 5-fold cross-validation.

Table 9 reports, for each (S, R) configuration, the smallest cross-validated RMSE over all 480 combinations of $(h, \bar{h})$. The selected configurations are $S^{(1)} \times S^{(2)} \times S^{(3)} = 3 \times 4 \times 1$, $R = 1$, $h = 1.80$, and $\bar{h} = 0.25$ for CFP, and $6 \times 2 \times 1$, $R = 1$, $h = 0.85$, and $\bar{h} = 0.30$ for ROI. Their standardized-scale RMSE values are 0.9890 and 0.9620.

Table 9: Minimum RMSE by tensor partition and CP rank.

$S^{(1)} \times S^{(2)} \times S^{(3)}$	CFP				ROI			
	$R = 1$	$R = 2$	$R = 3$	$R = 4$	$R = 1$	$R = 2$	$R = 3$	$R = 4$
$2 \times 2 \times 1$	0.9958	0.9993	1.0068	1.0125	0.9793	0.9746	0.9942	0.9964
$2 \times 3 \times 1$	1.0032	1.0213	1.0207	—	0.9849	0.9942	1.0219	—
$2 \times 4 \times 1$	1.0096	1.0219	1.0212	—	0.9924	1.0089	1.0370	—
$2 \times 6 \times 1$	1.0204	1.0369	1.1059	—	1.0093	1.0410	1.0448	—
$3 \times 2 \times 1$	0.9931	0.9993	1.0247	—	0.9742	0.9936	1.0190	—
$3 \times 3 \times 1$	0.9964	1.0261	1.0383	1.0440	0.9782	0.9978	1.0563	1.0698
$3 \times 4 \times 1$	0.9890	1.0025	1.0547	—	1.0054	1.0521	1.0905	—
$3 \times 6 \times 1$	1.0087	1.0596	1.1367	—	1.0112	1.0890	1.1446	—
$4 \times 2 \times 1$	1.0055	1.0135	1.0326	—	0.9766	0.9885	1.0242	—
$4 \times 3 \times 1$	1.0054	1.0626	1.1094	—	0.9823	1.0246	1.0476	—
$4 \times 4 \times 1$	1.0131	1.0382	1.1292	1.1915	1.0023	1.0397	1.1207	1.1167
$4 \times 6 \times 1$	1.0363	1.1582	1.2898	—	0.9768	1.0598	1.1639	—
$6 \times 2 \times 1$	1.0160	1.0358	1.0586	—	0.9620	0.9792	1.0512	—
$6 \times 3 \times 1$	1.0231	1.0709	1.1856	—	0.9904	1.0589	1.1264	—
$6 \times 4 \times 1$	1.0231	1.0666	1.1431	—	1.0174	1.1024	1.2372	—
$6 \times 6 \times 1$	1.0701	1.3510	1.7346	2.2838	1.0444	1.1539	1.3007	1.7721
$8 \times 8 \times 1$	1.1220	1.7289	3.4191	5.0691	1.1446	1.4350	3.5168	4.7495

Figure 6 shows an image example under our chosen hyperparameter.

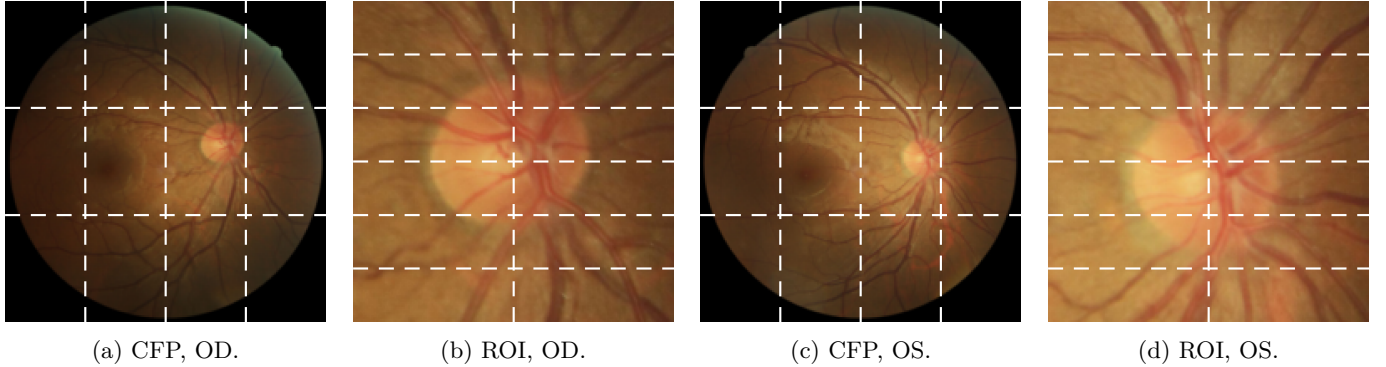


Figure 6: Representative paired-eye GRAPE images with partition $3 \times 4 \times 1$ for CFP and $6 \times 2 \times 1$ for ROI.

Model Interpretation 估计的 At 也可以画出来; 最后再做一个 bootstrap 来得到显著的 z 's! 图像可以参考 [VaryingCoefLM.pdf](#).

Paired vs iid VCTR Models Using the selected hyperparameters, we can compare the iid VCTR and paired VCTR models under the same 5-fold splits. Table 10 reports the mean and standard deviation of the RMSE across the five held-out folds. The paired VCTR achieves a lower mean RMSE than the iid VCTR for both image types. The improvement is more evident for CFP, where the mean RMSE decreases from 1.0163 to 0.9845. For ROI, the mean RMSE decreases slightly from 0.9610 to 0.9595.

Table 10: Fundus image analysis: RMSE for iid/paired VCTR models under 5-fold CV.

Model	CFP	ROI
VCTR (iid ver)	1.0163 ± 0.1491	0.9610 ± 0.0959
VCTR (paired ver)	0.9845 ± 0.1028	0.9595 ± 0.0904